

Calculus Lecture Note

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Remark 0.0.1. *Unless otherwise specified, all theorems, corollaries, propositions, remarks and problems are referring to [김홍중, 미적분학1 3rd ed.].*

1 1st week

1.1 What is “Function”?

Before discussing limits, we should first make precise what a *function* is. Imagine any **function**. What did you get? Something like a graph on $\mathbb{R}^2$ plane, or a formula like $y = f(x)$... right? Those are, in fact, not functions. Many highschoolers (or who were highschoolers until a few days ago) must have heard the definition of function, but since it is not empathized by their exams, so I guess that you may have forgot it.

Definition 1.1.1 (Function). *Let X and Y be sets. A map $f: X \rightarrow Y$ is called a **function** if each element $x \in X$ is assigned exactly one element of Y . In symbols,*

$$x \in X \quad \mapsto \quad f(x) \in Y.$$

The graph of a function can be written as the set

$$\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

This graph satisfies the following property:

$$(x_1, y_1), (x_2, y_2) \in \Gamma_f, \quad x_1 = x_2 \quad \implies \quad y_1 = y_2.$$

In other words, the same input cannot produce two different outputs.

1.2 Why Do We Need Limits?

The starting question of the lecture is the following:

“When x approaches a certain value, what does $f(x)$ approach?”

However, this question does not make sense in an arbitrary setting. If X and Y are merely sets, then expressions such as “approaches” or “is close to” are not yet meaningful. To talk about closeness, we need a notion of **distance** or **neighborhood**.

For that reason, calculus begins by working in the real numbers and considering situations such as

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(a) = b.$$

Intuitively, we want to express the idea that

if x is sufficiently close to a , then $f(x)$ is also close to b .

The goal is to formulate this statement with mathematical precision.

1.3 Closeness, Neighborhoods, and Absolute Value

Over the real numbers, closeness can be measured using absolute value.

- x is close to a if $|x - a|$ is small.
- $f(x)$ is close to b if $|f(x) - b|$ is small.

Let us first control the output. Choose any $\varepsilon > 0$ and require that

$$|f(x) - b| < \varepsilon.$$

Then the problem becomes:

Can we choose some $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - b| < \varepsilon$$

holds?

Here, δ measures the size of the input neighborhood, and ε measures the size of the output neighborhood.

1.4 The ε - δ Definition

Organizing the above intuition gives the standard definition of a limit.

Definition 1.4.1 (ε - δ Definition). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function, and let $a, b \in \mathbb{R}$. Then*

$$\lim_{x \rightarrow a} f(x) = b$$

means precisely the following:

For every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - b| < \varepsilon$$

holds.

Remark 1.4.2. *In the original motivation, one often imagines the case $f(a) = b$ and writes*

$$|x - a| < \delta \implies |f(x) - b| < \varepsilon.$$

*This is closer to the definition of **continuity**. In the standard definition of a limit, we usually exclude the point $x = a$ by writing $0 < |x - a| < \delta$.*

1.5 Properties of the Real Numbers: Completeness and the Archimedean Property

Mathematics begins with axioms, so I mention the following ‘property’. to completeness and the Archimedean property. One of the useful facts for limits and sequences is the following.

Proposition 1.5.1 (Archimedean Property). *For every real number $x \in \mathbb{R}$, there exists a natural number $n \in \mathbb{N}$ such that $n > x$.*

An equivalent formulation that is often used is

$$\forall \varepsilon > 0, \exists n \in \mathbb{N} \text{ such that } \frac{1}{n} < \varepsilon.$$

This fact is a basic tool in proving that $1/n \rightarrow 0$.

One may resolve oneself’s doubt toward remark 1.2.4 with the Archimedean property.

1.6 Problems

Problem 1.1.1. 1. Let $a > 1$. Compute

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^2}.$$

2. Let $a > 0$ and $k \in \mathbb{N}$. Compute

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^k}.$$

Proof. 1. By the remark 1.2.4 in the book, and since $f(x) = x^2$ is continuous, we have:

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{\sqrt{a}}{n} \right)^2 = 0.$$

2. If $a > 1$, we apply the same logic with (1). If $a = 1$, what we need to calculate is $\lim_{n \rightarrow \infty} \frac{1}{n^k}$. We claim that the answer is zero. By the Archimedean property, we have:

$$\begin{aligned} \exists N \in \mathbb{N} \text{ such that } n > N \Rightarrow 1 < n\epsilon^{\frac{1}{k}} &\iff 1 < n^k\epsilon \\ \iff \frac{1}{n^k} < \epsilon &\iff \left| \frac{1}{n^k} - 0 \right| < \epsilon \end{aligned}$$

Finally, if $0 < a < 1$, then clearly $0 < \frac{a^n}{n^k} < \frac{1}{n^k}$, so we may apply the squeeze theorem.

□

Problem 1.1.2. *Compute*

$$\lim_{n \rightarrow \infty} n^{1/n}.$$

Proof. Fix any $\epsilon > 0$. By the remark 1.2.4, we know that for sufficiently large n ,

$$n \leq 1 + n\epsilon + \frac{1}{2}n(n-1)\epsilon^2 < 1 + n\epsilon + \frac{1}{2}n(n-1)\epsilon^2 + \cdots = (1 + \epsilon)^n.$$

Therefore,

$$n \gg 1 \Rightarrow 1 + \epsilon \geq n^{\frac{1}{n}} \geq 1$$

Finally we obtain

$$n \gg 1 \Rightarrow \left| n^{\frac{1}{n}} \right| \leq \epsilon$$

Since ϵ was arbitrary selected, we get $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$. □

Problem 1.1.3. For $k \in \mathbb{N}$, compute

$$\lim_{n \rightarrow \infty} \frac{(\log n)^k}{n}.$$

Proof. Let $m = n^{1/k}$. Then $n = m^k$ and $\log n = k \log m$, so

$$\frac{(\log n)^k}{n} = \frac{(k \log m)^k}{m^k} = k^k \left(\frac{\log m}{m} \right)^k.$$

Using basic calculus, we have $\lim_{n \rightarrow \infty} \log x/x \rightarrow 0$, hence we obtain

$$\frac{(\log n)^k}{n} \rightarrow 0.$$

□

Problem 1.1.6. Determine whether each of the following series converges:

$$(1) \quad \sum_{n=1}^{\infty} \frac{n^2}{50n^2 + 1},$$

$$(2) \quad \sum_{n=1}^{\infty} \frac{2^n - 2^{-n}}{2^n + 2^{-n}},$$

$$(3) \quad \sum_{n=1}^{\infty} n \sin \frac{1}{n}.$$

Proof. Use the term test. All of them diverge. □

Problem 1.2.7. Assume $0 < \varepsilon < 1$. Show that

$$\left| \frac{1}{1 + \varepsilon} - (1 - \varepsilon + \varepsilon^2) \right| < \varepsilon^3.$$

Proof.

$$\begin{aligned} & \left| \frac{1}{1 + \varepsilon} - (1 - \varepsilon + \varepsilon^2) \right| \\ &= \left| \frac{1}{1 + \varepsilon} - \frac{1 + \varepsilon^3}{1 + \varepsilon} \right| \\ &= \frac{\varepsilon^3}{1 + \varepsilon} < \varepsilon^3 \end{aligned}$$

□

2 2nd week

2.1 Review

Remark 2.1.1. Be careful for the “positive sequence” condition of convergence tests.

Theorem (Comparison Test). Let $0 \leq a_n \leq b_n$ for all n .

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Remark 2.1.2. We can ignore finitely many terms when we determine whether a series converges or not. That is, finitely many terms cannot affect the convergence of a series.

By using the remark above, we can prove the following corollary:

Corollary (Generalized Limit Comparison Test). Let $a_n, b_n > 0$, and suppose

$$\lim_{n \rightarrow \infty} \frac{b_n}{a_n} = L, \quad L > 0.$$

Then $\sum a_n$ and $\sum b_n$ have the same behavior: they either both converge or both diverge.

Proof. Choose $0 < \varepsilon < \frac{1}{2}L$.

By the condition, $\exists N \in \mathbb{N}$ such that $n > N \Rightarrow \left| \frac{b_n}{a_n} - L \right| < \varepsilon$.

$$\therefore n > N \Rightarrow \frac{1}{2}L < \frac{b_n}{a_n} < 2L \Rightarrow \frac{1}{2}La_n < b_n < 2La_n.$$

When $n > N$, since $\frac{1}{2}La_n < b_n < 2La_n$, $\sum_{n=N+1}^{\infty} \frac{1}{2}La_n < \sum_{n=N+1}^{\infty} b_n < \sum_{n=N+1}^{\infty} 2La_n$.

1. $\sum_{n=1}^{\infty} b_n = \infty \iff \sum_{n=N+1}^{\infty} b_n = \infty \Rightarrow 2L \sum_{n=N+1}^{\infty} a_n = \infty \iff \sum_{n=1}^{\infty} a_n = \infty$.
2. $\sum_{n=1}^{\infty} b_n < \infty \iff \sum_{n=N+1}^{\infty} b_n < \infty \Rightarrow \frac{1}{2}L \sum_{n=N+1}^{\infty} a_n < \infty \iff \sum_{n=1}^{\infty} a_n < \infty$.
3. $\sum_{n=1}^{\infty} a_n = \infty \iff \frac{1}{2}L \sum_{n=N+1}^{\infty} a_n = \infty \Rightarrow \sum_{n=N+1}^{\infty} b_n = \infty \iff \sum_{n=1}^{\infty} b_n = \infty$.
4. $\sum_{n=1}^{\infty} a_n < \infty \iff 2L \sum_{n=N+1}^{\infty} a_n < \infty \Rightarrow \sum_{n=N+1}^{\infty} b_n < \infty \iff \sum_{n=1}^{\infty} b_n < \infty$.

□

Corollary (Root Test). In a positive-term series $\sum a_n$, if $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r$, then

1. $r < 1 : \sum a_n < \infty$.

2. $r > 1$: $\sum a_n = \infty$.

But what happens when $r = 1$?

ex 1: $a_n = n$, $\lim_{n \rightarrow \infty} n^{1/n} = 1$, but $\sum n = \infty$.

ex 2: $a_n = \frac{1}{n^2}$, $\lim_{n \rightarrow \infty} n^{-2/n} = 1$, but $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$.

Corollary (Ratio Test). *In a positive sequence (a_n) , if $\exists \rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$, then*

1. $\rho < 1 \Rightarrow \sum a_n < \infty$.

2. $\rho > 1 \Rightarrow \sum a_n = \infty$.

What happens when $\rho = 1$? The same examples as the root test show that the ratio test is also inconclusive when $\rho = 1$.

Remark 2.1.3. *If $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$ and $\lim_{n \rightarrow \infty} \sqrt[n]{a_n}$ both exist, in fact the two values are the same. In this case the two statements are equivalent. In general, the Root Test is the stronger statement than the Ratio Test.*

2.2 Problems

Problem 1.3.2. Show that every harmonic series diverges. That is, for given real numbers a, d , if a is not a natural number multiple of $-d$, show that the series

$$\sum_{n=1}^{\infty} \frac{1}{a + nd}$$

diverges.

Proof. WLOG, WMA $d = 1$. ($\because a \neq k(-d)$, $k \in \mathbb{N}$).

$$\text{WTS } \sum_{n=1}^{\infty} \frac{1}{a + n} = \infty.$$

$$\text{If } a \leq 0, \sum_{n=1}^{\infty} \frac{1}{n} < \sum_{n=1}^{\infty} \frac{1}{n + a}.$$

If $a > 0$, take $M = \min\{m \in \mathbb{N} \mid m > a\}$, possible by Archimedes' property.

$$\text{Then for all } n > M, \frac{1}{a + n} > \frac{1}{M + n}.$$

$$\therefore \sum_{n=M}^{\infty} \frac{1}{n} < \sum_{n=1}^{\infty} \frac{1}{n + a}.$$

□

Problem 1.3.3. Determine whether the following series converge.

$$1. \sum_{n=1}^{\infty} \frac{1}{\log(n+1)}$$

$$2. \sum_{n=1}^{\infty} \sin^2 \frac{\pi}{n}$$

$$3. \sum_{n=1}^{\infty} \frac{3n^2 + 4n + 5}{n^3 + 1}$$

$$4. \sum_{n=1}^{\infty} 2^{-n^2}$$

$$5. \sum_{n=1}^{\infty} \log \left(\frac{n^2 + 3n + 1}{n^2} \right)$$

$$6. \sum_{n=2}^{\infty} \frac{1}{(\log n)^{100}}$$

Proof.

$$(1) \frac{1}{n+1} < \frac{1}{\log(n+1)}$$

$$(2) \sin x < x \Rightarrow \sin^2 \frac{\pi}{n} < \frac{\pi^2}{n^2} \rightarrow \sum \sin^2 \frac{\pi}{n} < \pi^2 \sum \frac{1}{n^2} < \infty.$$

$$(3) \frac{3n^2}{n^3 + n^3} < \frac{3n^2 + 4n + 5}{n^3 + 1}$$

$$(4) 2^{-n^2} < 2^{-n}$$

$$(5) \frac{n^2 + 2n + 1}{n^2} < \frac{n^2 + 3n + 1}{n^2}$$

$$(6) \text{ For } n \gg 1, \frac{1}{n} < \frac{1}{(\log n)^{100}}.$$

□

Problem 1.3.5. (*Limit Comparison Test*) For positive sequences (a_n) and (b_n) , suppose $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$ (1)

Show that a necessary and sufficient condition for $\sum a_n$ to converge is that $\sum b_n$ converges.

What if the right-hand side of (1) is not 1 but 2?

Proof. Already Done.

□

Problem 1.4.1. Determine whether the following series converge.

$$(1) \sum \frac{2^n}{n^n}$$

$$(2) \sum \frac{2^n}{n^3}$$

$$\text{Proof. } (1) \lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^n}{n^n}} = \lim_{n \rightarrow \infty} \frac{2}{n} = 0 \quad \therefore \text{converges.}$$

$$(2) \lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^n}{n^3}} = \lim_{n \rightarrow \infty} \frac{2}{n^{3/n}} = 2. \quad \therefore \text{diverges.}$$

□

Problem 1.4.2. Find the range of positive a for which the series $\sum_{n=1}^{\infty} \left(\frac{an}{n+2} \right)^n$ converges.

$$\text{Proof. } \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{an}{n+2} \right)^n} = \lim_{n \rightarrow \infty} \frac{an}{n+2} = a$$

$\therefore 0 < a < 1$: converges & $a > 1$ diverges.

Suppose $a = 1$:

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+2} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+2}{n} \right)^n} = \lim_{n \rightarrow \infty} \frac{1}{\left(\left(1 + \frac{2}{n} \right)^{n/2} \right)^2} = \frac{1}{e^2} \neq 0.$$

$\therefore$ diverges by the divergence test.

□

Problem 1.4.3. Determine whether the following series converge.

$$(1) \sum_{n=1}^{\infty} \left(2 - \frac{n}{n+1}\right)^{n^2}$$

$$(2) \sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^{n^3}$$

$$(3) \sum_{n=1}^{\infty} \frac{1}{5^n} \left(1 + \frac{1}{n^2}\right)^{n^3}$$

$$(4) \sum_{n=1}^{\infty} \left(1 + \frac{1}{n} + \frac{1}{n^2}\right)^{-n^2}$$

$$(5) \sum_{n=1}^{\infty} \sin \left(\left(1 - \frac{1}{n^2}\right)^{n^3} \right)$$

Proof. (1) $\lim_{n \rightarrow \infty} \sqrt[n]{\left(2 - \frac{n}{n+1}\right)^{n^2}} = \lim_{n \rightarrow \infty} \left(2 - \frac{n}{n+1}\right)^n = \lim_{n \rightarrow \infty} \left(\left(1 + \frac{1}{n+1}\right)^{n+1} \right)^{\frac{n}{n+1}} = e$

$\therefore$ diverges.

(2) The logic that “ $0 < \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{n^2} < \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$, hence converges” is mistake! Be careful for the “oscillating sequences”.

But $\sum \left(1 - \frac{1}{n}\right)^{n^3} < \sum \left(1 - \frac{1}{n}\right)^{n^2} < \infty$ holds.

$$(3) \lim_{n \rightarrow \infty} \frac{1}{5} \left(1 + \frac{1}{n^2}\right)^{n^2} = \frac{e}{5} < 1.$$

$\therefore$ converges.

$$(4) \sum \left(1 + \frac{1}{n} + \frac{1}{n^2}\right)^{-n^2} < \sum \left(1 + \frac{1}{n}\right)^{-n^2}, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-n} = \frac{1}{e}.$$

$$(5) \sum \sin \left(\left(1 - \frac{1}{n^2}\right)^{n^3} \right) < \sum \left(1 - \frac{1}{n^2}\right)^{n^3}, \quad \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n^2}\right)^{n^2} = \frac{1}{e}.$$

□

Problem 1.5.1. Determine whether the following series converge or diverge.

$$(1) \sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$$

$$(2) \sum_{n=1}^{\infty} \frac{n^2}{e^n}$$

$$(3) \sum_{n=1}^{\infty} \frac{2^n}{n^2}$$

$$(4) \sum_{n=1}^{\infty} \frac{2^n}{n^3}$$

Proof. Write all the sequences as a_n .

$$(1) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} 2(n+1) \frac{n^n}{(n+1)^{n+1}} = \lim_{n \rightarrow \infty} 2 \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} 2 \left(1 + \frac{1}{n} \right)^{-n} = \frac{2}{e} < 1.$$

$$(2) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{e} \cdot \frac{(n+1)^2}{n^2} = \frac{1}{e} < 1.$$

$$(3) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} 2 \cdot \frac{n^2}{(n+1)^2} = 2 > 1.$$

$$(4) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n^3}{(n+1)^3} \cdot 2 = 2 > 1.$$

□

Problem 1.5.3. Find the range of x for which the following series converge.

$$(1) \sum_{n=1}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$(2) \sum_{n=1}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

Proof.

$$(1) \lim_{n \rightarrow \infty} \frac{x^2}{(2n+1)(2n+2)} = 0, \quad \text{converges for } x \in \mathbb{R}.$$

$$(2) \lim_{n \rightarrow \infty} \frac{x^2}{(2n+2)(2n+3)} = 0, \quad \text{converges for } x \in \mathbb{R}.$$

$$\text{In fact, } \cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

□

Problem 1.5.4. Determine whether the following series converge or diverge.

$$(1) \sum_{n=1}^{\infty} \frac{n \log n}{2^n}$$

$$(2) \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

$$(3) \sum_{n=1}^{\infty} \frac{n! \cdot n!}{(2n)!}$$

$$(4) \sum_{n=1}^{\infty} \frac{n^n}{(2n)!e^n}$$

$$(5) \sum_{n=1}^{\infty} \frac{n^n}{(2.7)^n n!}$$

$$(6) \sum_{n=1}^{\infty} a_n, \text{ where } a_1 = 1, a_{n+1} = \frac{2 + \cos n}{\sqrt{n}} a_n$$

$$(7) \sum_{n=1}^{\infty} \frac{n! \cdot 2^n}{(n + \sqrt{n})^n}$$

Proof.

$$(1) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{(n+1) \log(n+1)}{n \log n} = \frac{1}{2} < 1$$

$$(2) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0 < 1$$

$$(3) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)(n+1)}{(2n+2)(2n+1)} = \frac{1}{4} < 1$$

$$(4) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} \cdot \frac{1}{(2n+1)(2n+2)e} = 0 < 1$$

$$(5) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n^n} \cdot \frac{1}{(2.7)(n+1)} = \lim_{n \rightarrow \infty} \frac{1}{2.7} \left(1 + \frac{1}{n}\right)^n = \frac{e}{2.7} > 1, \quad \text{diverges.}$$

$$(6) 0 < \frac{a_{n+1}}{a_n} = \frac{2 + \cos n}{\sqrt{n}} < \frac{3}{\sqrt{n}}$$

$$\therefore 0 \leq \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2 + \cos n}{\sqrt{n}} \leq \lim_{n \rightarrow \infty} \frac{3}{\sqrt{n}} = 0 < 1.$$

$$(7) \text{ Note } 0 < \frac{n! \cdot 2^n}{(n + \sqrt{n})^n} < \frac{n! \cdot 2^n}{n^n}$$

$$\text{By 1.5.1.(1), } \sum \frac{n! \cdot 2^n}{n^n} < \infty.$$

$\therefore$ converges by Comparison Test.

□

3 3rd week

3.1 Review

We treated and will treat the following tests in this semester:

- Term Test(일반항 판정법); $T.T$
- Root Test(근 판정법); RoT
- Ratio Test(비 판정법); RaT
- Integral Test(적분 판정법); $I.T$
- Alternating Series Test(교대 급수 판정법); $A.S.T$
- Comparison Test(비교 판정법); $C.T$
- Limit Comparison Test(극한 비교 판정법); $L.C.T$

Lemma. For $s \in \mathbb{R}$,

$$\sum_{n=1}^{\infty} \frac{1}{n^s}$$

converges if and only if $s > 1$.

Proof. If $s \leq 1$, then $1/n^s \geq 1/n$ for all n , so the series diverges by comparison with the harmonic series. If $s > 1$, the series converges by the integral test. $\square$

3.1.1 Checklist for Integral Test

1. For sufficiently large x , is $f(x)$ positive?
2. For sufficiently large x , is $f(x)$ decreasing?

3.1.2 Checklist for Alternating Series Test

1. Is the series alternating?
2. Is the absolute value of the terms decreasing?
3. Does the absolute of the terms approach zero?

Bonus: Does the series absolutely converge?

3.2 Problems

Problem 1.6.1. *Decide whether each of the following series converges:*

$$\begin{array}{ll}
 (1) \sum_{n=1}^{\infty} \frac{-2}{n\sqrt{n}}, & (2) \sum_{n=1}^{\infty} \frac{\log n}{n}, \\
 (3) \sum_{n=1}^{\infty} \frac{\log n}{\sqrt{n}}, & (4) \sum_{n=1}^{\infty} \frac{2^n}{n+1}, \\
 (5) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}, & (6) \sum_{n=1}^{\infty} \frac{\sqrt{n}}{\log(n+1)}, \\
 (7) \sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n, & (8) \sum_{n=1}^{\infty} \frac{1}{(\log 2)^n}, \\
 (9) \sum_{n=1}^{\infty} \frac{1}{(\log 3)^n}, & (10) \sum_{n=1}^{\infty} n \tan \frac{1}{n}.
 \end{array}$$

Proof.

1. Clear by the lemma.
2. Diverges by the integral test.
3. Diverges by comparison with (2), because $\sqrt{n} < n$ gives $\frac{\log n}{\sqrt{n}} > \frac{\log n}{n}$.
4. Diverges by the term test, since $\frac{2^n}{n+1} \not\rightarrow 0$.
5. Diverges by limit comparison with $1/n$:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{\sqrt{n}(\sqrt{n}+1)}}{1/n} = 1.$$

6. Diverges by comparison, because for large n one has

$$\frac{1}{n} < \frac{\sqrt{n}}{\log(n+1)}.$$

7. Diverges by the term test, since $\left(1 + \frac{1}{n}\right)^n \rightarrow e$.
8. Diverges: $\sum \frac{1}{r^n} = \frac{1}{1-r}$ iff $|r| > 1$. $0 < \log 2 < 1$.
9. Converges: $\log 3 > 1$.
10. Diverges by the term test. $\lim_{n \rightarrow \infty} n \tan \frac{1}{n} = 1$.

□

Problem 1.6.3. Use the integral test to show the following:

$$\begin{aligned} (1) \quad & \sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1+\varepsilon}} < \infty \quad (\varepsilon > 0), \\ (2) \quad & \sum_{n=3}^{\infty} \frac{1}{n \log n \log \log n} = \infty, \\ (3) \quad & \sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^{1+\varepsilon}} < \infty \quad (\varepsilon > 0). \end{aligned}$$

Proof.

1. For $x \gg 2$,

$$f(x) := \frac{1}{x(\log x)^{1+\varepsilon}} > 0, \quad f'(x) = -\frac{1}{x^2(\log x)^{1+\varepsilon}} - \frac{(1+\varepsilon)}{x^2(\log x)^{2+\varepsilon}} < 0.$$

$$\lim_{n \rightarrow \infty} \int_2^n \frac{1}{x(\log x)^{1+\varepsilon}} dx = \lim_{n \rightarrow \infty} \left[-\frac{1}{\varepsilon} \frac{1}{(\log x)^\varepsilon} \right]_2^n = \lim_{n \rightarrow \infty} \frac{1}{\varepsilon} \frac{1}{(\log 2)^\varepsilon} - \frac{1}{\varepsilon} \frac{1}{(\log n)^\varepsilon} = \frac{1}{\varepsilon} \frac{1}{(\log 2)^\varepsilon}.$$

$$\therefore \sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1+\varepsilon}} < \infty.$$

2. For $x \gg 3$,

$$f(x) := \frac{1}{x \log x \log \log x} > 0, \quad f'(x) = -\frac{\log x \log \log x + \log \log x + 1}{(x \log x \log \log x)^2} < 0.$$

$$\lim_{n \rightarrow \infty} \int_3^n \frac{1}{x \log x \log \log x} dx = \lim_{n \rightarrow \infty} [\log \log \log x]_3^n = \infty \iff \sum_{n=3}^{\infty} \frac{1}{n \log n \log \log n} = \infty.$$

3. For $x \gg 3$,

$$f(x) := \frac{1}{x \log x (\log \log x)^{1+\varepsilon}} > 0, \quad f'(x) = -\frac{\{\log((\log x) \cdot ((\log x + 1) \log \log x + 1 + \varepsilon))\}^{-2-\varepsilon}}{x^2(\log x)^2} < 0.$$

$$\lim_{n \rightarrow \infty} \int_3^\infty \frac{1}{x \log x (\log \log x)^{1+\varepsilon}} dx = \lim_{n \rightarrow \infty} \left[-\frac{1}{\varepsilon} \frac{1}{(\log \log x)^\varepsilon} \right]_3^n = \frac{1}{\varepsilon} \frac{1}{(\log \log 3)^\varepsilon}$$

$$\therefore \sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^{1+\varepsilon}} < \infty.$$

□

Problem 1.6.5. Find all real numbers p for which

$$\sum_{n=1}^{\infty} \frac{\log n}{n^p}$$

converges.

Proof. If $p \leq 0$, it diverges because of the comparison test.

$$\begin{aligned} f(x) &:= \frac{\log x}{x^p} > 0, & (x > 3) \\ f'(x) &= \frac{1 - p \log x}{x^{p+1}} < 0 & \left(x > e^{\frac{1}{p}}\right). \end{aligned}$$

Take $N = \min \left\{ n \in \mathbb{Z} : n > 3 \text{ and } n > e^{\frac{1}{p}} \right\}$.

$$\begin{aligned} \int_N^\infty \frac{\log x}{x^p} dx &= \int_{\log N}^\infty \frac{1}{e^{(p-1)t}} dt \\ &= \begin{cases} \left[\frac{e^{(1-p)t}}{1-p} \right]_{\log N}^\infty & \cdot : \begin{cases} p < 1 \rightarrow \infty, \\ p > 1 \rightarrow \frac{1}{(p-1)e^{(p-1)\log N}}, \end{cases} \\ \int_{\log N}^\infty 1 dt \rightarrow \infty & p = 1. \end{cases} \end{aligned}$$

$$\therefore \sum \frac{\log n}{n^p} \text{ conv} \iff p > 1. \quad \square$$

Problem 1.6.7. Show that

1. $\int_1^\infty \sin \frac{1}{\sqrt{x}} dx = \infty$
2. $\int_1^\infty \frac{1}{x} \sin \frac{1}{x} dx < \infty$

Proof.

1.

$$\sum_{n=1}^\infty \frac{1}{\sqrt{n}} = \infty \xLeftrightarrow{\text{L.C.T.}} \sum_{n=1}^\infty \sin \frac{1}{\sqrt{n}} = \infty \xLeftrightarrow{\text{I.T.}} \int_1^\infty \sin \frac{1}{\sqrt{x}} dx = \infty.$$

2.

$$\sum_{n=1}^\infty \frac{1}{n^2} < \infty \xLeftrightarrow{\text{L.C.T.}} \sum_{n=1}^\infty \frac{1}{n} \sin \frac{1}{n} < \infty \xLeftrightarrow{\text{I.T.}} \int_1^\infty \frac{1}{x} \sin \frac{1}{x} dx < \infty. \quad \square$$

Problem 1.7.1. Classify the following series as absolutely convergent, conditionally convergent, or divergent:

- | | |
|--|---|
| (1) $\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}+1}{n+1},$ | (2) $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n},$ |
| (3) $\sum_{n=1}^{\infty} (-1)^n \frac{n}{e^n},$ | (4) $\sum_{n=1}^{\infty} (-1)^n \sin \frac{1}{n},$ |
| (5) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^{3/2}},$ | (6) $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n}{10}\right)^n,$ |
| (7) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{10^n}{n^{10}},$ | (8) $\sum_{n=1}^{\infty} (-1)^n \frac{\log n}{n},$ |
| (9) $\sum_{n=1}^{\infty} (-1)^n \log\left(1 + \frac{1}{n}\right),$ | (10) $\sum_{n=2}^{\infty} (-1)^n \frac{1}{\log n}.$ |

Proof.

1. A.C: $\times$ by L.C.T.

(a) OK,

(b) OK,

$$(c) f(x) = \frac{\sqrt{x}+1}{x+1}, f'(x) = \frac{\frac{x+1}{2\sqrt{x}} - \sqrt{x} - 1}{(x+1)^2} = \frac{-x - 2\sqrt{x} + 1}{(x+1)^2 2\sqrt{x}} < 0, \text{ OK.}$$

$\therefore$ Converges by A.S.T.

2. Note, $\cos n\pi = (-1)^n$. A.C: $\times$

(a) OK,

(b) OK,

(c) OK.

$\therefore$ converges by A.S.T.

□

$$\text{then } \sum_{n=1}^{\infty} a_n - \sum_{k=1}^n a_k < |a_{n+1}|.$$
$$\text{WTS } |S - S_n| \leq |S_{n+1} - S_n|.$$

Define $S_n := \sum_{k=1}^n a_k$. Note $a_{2m} + a_{2m+1} < 0$, and $a_{2m-1} + a_{2m} > 0$.

$$a_1 \geq a_1 + \underset{S_3}{(a_2 + a_3)} \geq a_1 + \underset{S_5}{(a_2 + a_3)} + (a_4 + a_5) + \cdots \geq \cdots \geq \underset{S_1}{S_{2n-1}} \quad \vee \quad (\because a_{2n})$$

$$a_1 + a_2 \leq (a_1 + a_2) + (a_3 + a_4) \leq (a_1 + a_2) + (a_3 + a_4) + (a_5 + a_6) + \cdots \leq S_{2n}$$

$$\therefore S_2 \leq S_4 \leq \cdots \leq S \leq \cdots \leq S_3 \leq S_1.$$

Problem 1.7.5. Approximate each quantity with error less than 10^{-2} :

- (1) $\log 1.1 = \frac{1}{10} - \frac{1}{2 \cdot 10^2} + \frac{1}{3 \cdot 10^3} - \cdots,$
- (2) $\sin 1 = 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \cdots,$
- (3) $\int_0^1 e^{-x^2} dx = 1 - \frac{1}{1! \cdot 3} + \frac{1}{2! \cdot 5} - \frac{1}{3! \cdot 7} + \cdots,$
- (4) $\int_0^1 \frac{\sin x}{x} dx = 1 - \frac{1}{3! \cdot 3} + \frac{1}{5! \cdot 5} - \frac{1}{7! \cdot 7} + \cdots.$

Proof. Use the previous problem. Approximate each quantity with error less than 10^{-2} :

$$1. \frac{1}{2 \cdot 10^2} < 10^{-2} \rightarrow \left| \log 1.1 - \frac{1}{10} \right| < 10^{-2}.$$

$$2. \quad \frac{1}{5!} < 10^{-2} \rightarrow \left| \sin 1 - \frac{5}{6} \right| < 10^{-2}.$$

$$3. \frac{1}{4! \cdot 9} < 10^{-2} \rightarrow \left| \int_0^1 e^{-x^2} dx - \left(1 - \frac{1}{1! \cdot 3} + \frac{1}{2! \cdot 5} - \frac{1}{3! \cdot 7} \right) \right| < 10^{-2}.$$

$$4. \frac{1}{5! \cdot 5} < 10^{-2} \rightarrow \left| \int_0^1 \frac{\sin x}{x} dx - \frac{17}{18} \right| < 10^{-2}.$$

Problem 2.2.1. Find the range of x for which each of the following series converges.

$$1. \sum \frac{x^n}{n^n}$$

$$2. \sum \frac{x^n}{\log(n+2)}$$

$$3. \sum \frac{nx^n}{6^n}$$

$$4. \sum \frac{n^2 x^n}{1+n^2}$$

$$5. \sum \frac{7^n x^n}{n!}$$

$$6. \sum (-1)^n \frac{(x-2)^n}{n^n}$$

$$7. \sum (\log n) x^n$$

$$8. \sum \frac{(x-1)^n}{(\log(n+1))^n}$$

$$9. \sum \frac{n(n+1)(n+2)}{3^n} x^n$$

$$10. \sum \left(n \sin \frac{1}{n} \right) x^n$$

$$11. \sum \left(n \tan \frac{1}{n} \right) x^n$$

$$12. \sum \left(\sin \frac{1}{n} \right) x^n$$

$$13. \sum \left(\tan \frac{1}{n} \right) x^n$$

$$14. \sum \left(\frac{1}{n} \tan \frac{1}{n} \right) x^n$$

$$15. \sum \frac{x^n}{\log(n+1)}$$

$$16. \sum \frac{x^n}{n2^n}$$

$$17. \sum \frac{x^n}{2^{n+(-1)^n}}$$

$$18. \sum_{n \geq 2} \frac{x^n}{3^{\log n} - 2^{\log n}}$$

$$19. \sum \frac{\sin(n\pi/2)}{2^n} x^n$$

$$20. \sum \frac{\cos n\pi}{n} x^n$$

Proof. 1. $\sum \frac{|x|^n}{n^n} < \infty$ for all $x \in \mathbb{R}$ by the Root Test.

Hence $\sum \frac{x^n}{n^n} < \infty$ for all $x \in \mathbb{R}$ by absolute convergence.

2. (a) If $|x| > 1$, then it diverges by the term test.

(b) If $x = 0$, then it is clear.

(c) If $0 < x < 1$, then it converges since $\sum \frac{x^n}{\log(n+2)} < \sum x^n$.

(d) If $-1 \leq x < 0$, then it converges by the alternating series test.

(e) If $x = 1$, then it diverges since $\sum \frac{1}{n+2} < \sum \frac{1}{\log(n+2)}$.

□

Let's stop here for a second. Isn't this process too boring? For me, yes it is. But fortunately, we can simplify our answers;

Theorem. Let $f(x) = \sum a_i x^i$. If f converges at $x = x_0$ and diverges at $x = x_1$, then:

1. $f(x)$ absolutely converges on $|x| < |x_0|$.

2. $f(x)$ diverges on $|x| > |x_1|$.

Indeed, the theorem above shows the existence of the radius of convergence. In particular, it gives us symmetry; thus we can focus on the positive axis when we determine the radius of convergence. Let's see the problem again:

Solution of Problem 2.2.1. By the theorem above, we may assume that $x \geq 0$.

$$(2) \sum \frac{x^n}{\log(n+2)} < \sum x^n$$

$$\rightarrow |x| < 1 \text{ conv.}$$

$$\& x = -1 \text{ conv.}$$

$$\rightarrow -1 \leq x < 1$$

$$(3) \text{ Ro.T: } \lim_{n \rightarrow \infty} \left| n^{1/n} \cdot \frac{x}{6} \right| = \frac{x}{6}.$$

$$\therefore |x| < 6 \text{ conv.}$$

If $x = 6$, clearly div.

$$x = -6, \text{ div } \because \text{T.T.}$$

$$\rightarrow -6 < x < 6$$

$$(4) \text{ By L.C.T., } \sum \frac{n^2}{1+n^2} x^n < \infty \iff \sum x^n < \infty.$$

$$\therefore |x| < 1 \text{ conv.}$$

$$x = \pm 1 \text{ div } \because \text{T.T.}$$

$$\rightarrow -1 < x < 1$$

$$(5) \text{ Ra.T: } \lim_{n \rightarrow \infty} \left| \frac{7^{n+1}x^{n+1}}{(n+1)!} \cdot \frac{n!}{7^n \cdot x^n} \right| = \lim_{n \rightarrow \infty} \frac{7x}{n+1} = 0 \quad \forall x.$$

$\therefore \text{ conv } \forall x \in \mathbb{R}$

$(-\infty, \infty)$

$$(6) \sum \left| (-1)^n \frac{(x-2)^n}{n^n} \right| = \sum \frac{(x-2)^n}{n^n}$$

$$\text{Ro.T } \rightarrow \lim_{n \rightarrow \infty} \left| \frac{x-2}{n} \right| = 0 \quad \forall x \in \mathbb{R}.$$

$$\therefore \sum (-1)^n \frac{(x-2)^n}{n^n} < \infty \quad \forall x \in \mathbb{R} \quad \therefore \text{ A.C.}$$

$(-\infty, \infty)$

$$(7) \text{ Ra.T: } \lim_{n \rightarrow \infty} \left| \frac{\log(n+1)x^{n+1}}{(\log n)x^n} \right| = x.$$

$\therefore |x| < 1 \text{ conv.}$

$x = \pm 1$ clearly div. ($\therefore$ T.T.)

$\rightarrow -1 < x < 1$

(8) Ro.T

$$\lim_{n \rightarrow \infty} \frac{x-1}{\log(n+1)} = 0 \quad \forall x \in \mathbb{R}.$$

$(-\infty, \infty)$

(9) Ra.T

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{x}{3}, \quad \therefore |x| < 3$$

$x = \pm 3 \dots$ clearly div. (T.T.)

$\rightarrow -3 < x < 3$

$$(10),(11) \sum \left(n \tan \frac{1}{n} \right) x^n \Longleftrightarrow \sum \left(n \sin \frac{1}{n} \right) x^n \Longleftrightarrow \sum x^n ; \quad \therefore \text{ L.C.T.}$$

$|x| < 1 \text{ conv.}$

$\rightarrow -1 < x < 1$

$$(12),(13) \sum \left(\tan \frac{1}{n} \right) x^n \Longleftrightarrow \sum \left(\sin \frac{1}{n} \right) x^n \Longleftrightarrow \sum \frac{x^n}{n} ; \quad \therefore \text{ L.C.T.}$$

$$\text{Ro.T } \lim_{n \rightarrow \infty} \left| \frac{x}{n^{1/n}} \right| = x \rightarrow |x| < 1 \text{ conv.}$$

$x = 1 \text{ div.}$

$x = -1 \text{ conv } \therefore \text{ A.S.T.}$

$\rightarrow -1 \leq x < 1$

$$(14) \sum \left(\frac{1}{n} \tan \frac{1}{n} \right) x^n \Longleftrightarrow \sum \frac{x^n}{n^2}$$

$$\text{Ro.T } \lim_{n \rightarrow \infty} \left| \frac{x}{n^{2/n}} \right| = x \rightarrow |x| < 1 \text{ conv, } x = 1 \text{ conv } \therefore \text{ Lemma (or I.T)}$$

$x = -1 \text{ conv } \therefore \text{ A.S.T.}$

$\rightarrow -1 \leq x \leq 1$

$$(15) \text{ Ra.T: } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = x.$$

$$\therefore |x| < 1 \text{ conv.}$$

$$x = 1 \text{ div } \because \frac{1}{n+1} < \frac{1}{\log(nx)}$$

$$x = -1 \text{ conv } \because \text{A.S.T.}$$

$$\rightarrow -1 \leq x < 1$$

$$(16) \text{ Ro.T } \lim_{n \rightarrow \infty} \left| \frac{1}{n^{1/n}} \cdot \frac{x}{2} \right| = \frac{x}{2}, \quad \therefore |x| < 2 \text{ conv.}$$

$$x = 2 \text{ div } \because \text{Lemma}$$

$$x = -2 \text{ conv } \because \text{A.S.T.}$$

$$\rightarrow -2 \leq x < 2$$

$$(17) \text{ L.C.T: } \sum \frac{x^n}{2^{n+(-1)^n}} \iff \sum \frac{x^n}{2^n}$$

$$\rightarrow -2 < x < 2$$

$$(18) \text{ L.C.T: } \sum \frac{x^n}{3^{\log n} - 2^{\log n}} \iff \sum \frac{x^n}{3^{\log n}}$$

$$\text{Ro.T: } \lim_{n \rightarrow \infty} \left| \frac{x}{3^{\log n^{1/n}}} \right| = x. \therefore |x| < 1, \quad x = -1 : \text{conv } \because \text{A.S.T.}$$

$$x = 1 : \sum \frac{1}{3^{\log n}} = \sum \frac{1}{n^{\log 3}}, \quad \log 3 > 1, \quad \therefore \text{conv } \because \text{Lemma. (or I.T.)}$$

$$\rightarrow -1 \leq x \leq 1$$

$$(19) \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2}\right) \frac{x^n}{2^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^{2n-1}} x^{2n-1}$$

$$\rightarrow |x| < 2 \text{ conv } \because \text{A.C. \& Ra.T.}$$

$$x = \pm 2 \text{ div.}$$

$$\rightarrow -2 < x < 2$$

$$(20) \sum \frac{\cos n\pi}{n} x^n = \sum \frac{(-1)^n}{n} x^n.$$

$$\rightarrow |x| < 1 \text{ conv } \because \text{A.C.}, x = 1 \text{ conv } \because \text{A.S.T}, x = -1 \text{ div } \because \text{Lemma}$$

$$\rightarrow -1 < x \leq 1.$$

□

Problem 2.2.2. Find the range of x for which each of the following series converges.

$$1. \sum_{n \geq 2} (-1)^n \frac{(2x-1)^n}{4^n \log n}$$

$$2. \sum n x^{n^2}$$

Proof.

$$1. \sum \left| (-1)^n \frac{(2x-1)^n}{4^n \log n} \right| = \sum \frac{(2x-1)^n}{4^n \log n}.$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{2x-1}{4} \rightarrow -\frac{3}{2} < x < \frac{5}{2} \text{ conv.}$$

$$\text{If } x = -\frac{3}{2}, \sum \frac{1}{\log n} > \sum \frac{1}{n} = \infty.$$

$$\text{If } x = \frac{5}{2}, \sum \frac{(-1)^n}{\log n} < \infty \because \text{A.S.T.}$$

$$\therefore -\frac{3}{2} < x \leq \frac{5}{2}.$$

2. WMA $x > 0$.

$$(a) 0 < x < 1 \rightarrow \text{RoT } \lim_{n \rightarrow \infty} n^{1/n} \cdot x^n = 0 \rightarrow \text{conv.}$$

$$(b) x = 1 \rightarrow \sum n = \infty.$$

$$(c) x = -1 \rightarrow \sum (-1)^n \dots \text{div. } \because \text{T.T.}$$

$$\therefore -1 < x < 1.$$

□

Problem 2.2.3. Show that the following series converge and compute their sums.

$$1. \sum \frac{n}{4^n}$$

$$2. \sum \frac{n(n-1)}{5^n}$$

$$3. \sum \frac{n(n-1)(n-2)}{2^n}$$

$$4. \sum \frac{n^2}{2^n}$$

$$5. \sum \frac{1}{n2^n}$$

$$6. \sum \frac{1}{n(n+1)2^n}$$

$$7. \sum \frac{(-1)^{n-1}}{n} \frac{1}{3^n}$$

$$8. \sum \frac{(-1)^n n^2}{2^n}$$

Proof.

$$1. \text{ Recall } \frac{1}{1-x} = \sum x^n, -1 < x < 1.$$

$$\rightarrow \frac{1}{(1-x)^2} = \sum nx^{n-1} \rightarrow \frac{x}{(1-x)^2} = \sum nx^n, \text{ put } x = \frac{1}{4}.$$

$$\therefore \sum \frac{n}{4^n} = \frac{4}{9}.$$

2. Differentiate one more time, $\frac{2}{(1-x)^3} = \sum n(n-1)x^{n-2}$, multiply x^2 , put $x = \frac{1}{5}$,

$$\rightarrow \sum \frac{n(n-1)}{5^n} = \frac{5}{32}.$$

3. Diff one more time

$$\rightarrow \frac{6}{(1-x)^4} = \sum n(n-1)(n-2)x^{n-3}, \text{ mult } x^3 \text{ both side \& put } x = \frac{1}{2}.$$

$$\rightarrow \sum \frac{n(n-1)(n-2)}{2^n} = 12$$

4. From (1), (2), we got $\frac{x}{(1-x)^2} = \sum nx^n$, $\frac{2x^2}{(1-x)^3} = \sum n(n-1)x^n$.

$$\text{Add them up, } \frac{x}{(1-x)^2} + \frac{2x^2}{(1-x)^3} = \sum n^2x^n, \text{ put } x = \frac{1}{2}.$$

$$\therefore \sum \frac{n^2}{2^n} = 6.$$

5. $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$. Integrate both sides.

$$\therefore -\log(1-x) = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = \sum_{n=1}^{\infty} \frac{1}{n} x^n, \text{ put } x = \frac{1}{2}.$$

$$\therefore \log 2 = \sum_{n=1}^{\infty} \frac{1}{n} x^n.$$

6. Integrate once more!

$$\int_0^x -\log(1-x) = -(x-1)\log(1-x) + x = \sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} x^{n+2} = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^{n+1}$$

$$\text{Put } x = \frac{1}{2}, \text{ multiply 2. } \therefore \sum \frac{1}{n(n+1)2^n} = 1 - \log 2.$$

7. Put $x = -\frac{1}{3}$ in (5), and multiply -1 .

$$\therefore \log \frac{4}{3}.$$

8. Put $x = -\frac{1}{2}$ in (4).

$$\therefore -\frac{2}{27}$$

□

Problem 2.2.4. Let $f(x) = \log\left(\frac{1+x}{1-x}\right)$ for $|x| < 1$. Answer the following questions.

1. Express $f(x)$ as a power series.

2. Find an approximation of $\log 2$ such that the error does not exceed 10^{-3} .

Proof.

$$\begin{aligned}
1. \quad f(x) &= \log(1+x) - \log(1-x) \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} + \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} \\
&= 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \quad (|x| < 1)
\end{aligned}$$

$$2. \quad \frac{1+x}{1-x} = 2 \rightarrow x = \frac{1}{3}.$$

$$f\left(\frac{1}{3}\right) = 2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)3^{2n+1}}.$$

$$\text{Note } f\left(\frac{1}{3}\right) - \left(\frac{1}{3} + \frac{1}{3 \cdot 3^3} + \frac{1}{5 \cdot 3^5}\right) = 2 \sum_{n=3}^{\infty} \frac{1}{(2n+1)3^{2n+1}} < \frac{2}{3} \sum_{n=3}^{\infty} \frac{1}{n \cdot 9^n} = \frac{1}{3 \cdot 9^2 \cdot 4 \cdot 7} < 10^{-3}.$$

$$\therefore \log 2 \sim 2 \left(\frac{1}{3} + \frac{1}{3 \cdot 3^3} + \frac{1}{5 \cdot 3^5} \right).$$

□

Problem 2.2.5.

A power series need not be defined only near the origin. A power series defined near the point $x = c$ is expressed as $\sum_{n=0}^{\infty} a_n(x-c)^n$.

If the radius of convergence r of this power series is positive, then for every real number x with $|x-c| < r$, the function $f(x) := \sum_{n=0}^{\infty} a_n(x-c)^n$ is infinitely differentiable. Show that $a_n = \frac{f^{(n)}(c)}{n!}$.

Proof. Put $x-c = t$.

$$\therefore f(t+c) = g(t) = \sum_{n=0}^{\infty} a_n t^n.$$

$$\rightarrow a_n = \frac{g^{(n)}(0)}{n!} = \frac{f^{(n)}(c)}{n!}.$$

□

4 4th week

4.1 A Breif Recall of The Last Week

1. Term Test
2. Comparison Test
3. Limit Comparison Test
4. Root Test
5. Ratio Test
6. Integral Test
7. Alternating Series Test

During 3 weeks, we have learned about how to distinguish the given series is whether converging, or diverging. But, why? Why do we need those? As I said last week, our ultimate goal is to **approximate** non-elementary functions, such as $y = e^x$.

Last week, we have prepared to approximate such functions. The most important theorems we learned were following:

Theorem. *Suppose that the power series*

$$f(x) = \sum a_i x^i$$

converges at $x = x_0$, and diverges at $x = x_1$. Then:

1. $f(x)$ absolutely converges on $|x| < |x_0|$.
2. $f(x)$ diverges on $|x| > |x_1|$.

□

The following corollary was the main material we used last week.

Corollary. *Let the power series $f(x) = \sum a_i x^i$ be given. If $\forall a_i \neq 0$, and if*

$$\ell := \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \in [0, \infty]$$

*exists, then the **radius of convergence** of $f(x)$ is $\frac{1}{\ell}$.*

□

Also, the following theorem is the most powerful tool which admits us to approximate some non-elementary function.

Theorem (The Fundamental Theorem of Power Series). *Let the power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be given. Suppose that its radius of convergence is $r > 0$. Then the followings hold:*

1. The radius of convergence of $\sum_{n=1}^{\infty} na_n x^{n-1}$ is also r .

2. The radius of convergence of $\sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ is also r .

3. $f(x)$ is differentiable on $(-r, r)$, and in particular,

$$(a) f'(x) = \sum_{n=1}^{\infty} na_n x^{n-1} \text{ on } (-r, r)$$

$$(b) \int_0^x f(t)dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \text{ on } (-r, r)$$

4. If $f(x)$ is a power series,

$$a_n = \frac{f^{(n)}(0)}{n!}$$

holds.

□

We observed some basic examples, such as $f(x) = \frac{1}{1-x}$. Today, we are going to observe some more beautiful objects.

4.2 More Functions and Inverse Function Theorem

The following functions are the examples that we are going to use today. In fact, there are some more theories, but let's talk about it somewhat later.

$$1. e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$2. \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$3. \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$4. \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$5. \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

Also, we will calculate the derivative of the inverse function, using the following theorem.

Theorem (Inverse Function Theorem). *Let $f : I \rightarrow \mathbb{R}$ is differentiable, and suppose that for all $x \in I$, $f'(x) \neq 0$. Then*

1. f is bijective.

2. Image of f , i.e. $f(I)$ is open interval in $\mathbb{R}$.

3. The inverse function of f is well-defined. Write the inverse of f as f^{-1} .

4. $f^{-1} : f(I) \rightarrow I$ is differentiable.

5. Suppose $f(x_0) = y_0$. Then $\frac{d}{dy}f^{-1}(y_0) = \frac{1}{\frac{d}{dx}f(x_0)}$.

□

Note that the **inverse function** is the special case of the inverse image. The inverse image of f is defined by:

$$f^{-1}(\alpha) = \{x \in I : f(x) = \alpha\}.$$

Using this fact, we can calculate some more power series.

1. $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$

2. $\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$

To get precise power series, we need some calculations. This will be treated soon.

4.3 Problems

Problem 2.3.4. Express the following function as a power series centered at $x = 1$:

$$\int_1^x e^{-\frac{(t-1)^2}{2}} dt$$

Proof. Expansion of $y = e^x$ as a power series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (x \in \mathbb{R})$$

Substituting $x = -\frac{(t-1)^2}{2}$ gives:

$$e^{-\frac{(t-1)^2}{2}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n \cdot n!} (t-1)^{2n} \quad (t \in \mathbb{R})$$

Applying the FTPS, we get:

$$\int_1^x e^{-\frac{(t-1)^2}{2}} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n (2n+1) n!} (x-1)^{2n+1} \quad (x \in \mathbb{R})$$

□

Problem 2.3.7 (Binomial Expansion). For a given $r \in \mathbb{N}$, if we expand $(1+x)^r$, the coefficient of x^n is

$$\binom{r}{n} := \frac{r(r-1)\cdots(r-n+1)}{1 \cdot 2 \cdots n}$$

Therefore,

$$(1+x)^r = \sum_{n=0}^r \binom{r}{n} x^n$$

Even we can extend this to $r \in \mathbb{R}$. Note that this is defined for $n \in \mathbb{Z}_{\geq 0}$, where $\binom{r}{0} = 1$. If $r \in \mathbb{R} \setminus \mathbb{N}$, then the power series becomes

$$f(x) := \sum_{n=0}^{\infty} \binom{r}{n} x^n = 1 + rx + \frac{r(r-1)}{2} x^2 + \cdots$$

Show that the convergence range of $f(x)$ is 1, and $f(x)$ satisfies the following equation:

$$(1+x)f'(x) = rf(x) \quad (|x| < 1).$$

From this, show that the **Newton's binomial formula** holds:

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n \quad (|x| < 1)$$

Proof.

$$\lim_{n \rightarrow \infty} \left| \frac{\binom{r}{n+1}}{\binom{r}{n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{r-n}{n+1} \right| = 1$$

Therefore, the radius of convergence of the power series above is 1.

Also $(1+x)^r$ satisfies the differential equation above, by the Fundamental Theorem of Power Series.

Now the facts that the derivative of $F(x) = \frac{f(x)}{(1+x)^r}$ is constantly one, and $F(0) = 1$ finish the proof. $\square$

Problem 2.3.10. When $r > 0$, it is known that the product of two power series converging on $(-r, r)$ also converges on $(-r, r)$. Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ are given. Show that

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n \quad (|x| < r)$$

holds, by the following method.

1. (Leibniz Rule) Show that

$$\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x)$$

holds for f, g which are differentiable n times.

2. Show that the equation above holds

3. Express $y = e^{x+x^2}$ as a power series.

Proof.

1. Use induction. Clearly it holds for $n = 1$. Suppose it holds for n . Then:

$$\begin{aligned} \frac{d^{n+1}}{dx^{n+1}}(f(x)g(x)) &= \sum_{k=0}^n \binom{n}{k} f^{(k+1)}(x) g^{(n-k)}(x) + \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k+1)}(x) \\ &= \sum_{k=1}^{n+1} \binom{n}{k-1} f^{(k)}(x) g^{(n-k+1)}(x) + \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k+1)}(x) \\ &= f^{(n+1)}(x) g(x) + f(x) g^{(n+1)}(x) + \sum_{k=1}^n \binom{n+1}{k} f^{(k)}(x) g^{(n-k+1)}(x) \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k} f^{(k)}(x) g^{(n-k+1)}(x) \end{aligned}$$

2. Put $x = 0$ in (1). Use the fact that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Then

$$\frac{d^{n+1}}{dx^{n+1}}(f(x)g(x)) \big|_{x=0} = (n+1)! \sum_{k=0}^{n+1} \frac{f^{(k)}(0)}{k!} \frac{g^{(n-k+1)}(0)}{(n-k+1)!}$$

3. Consider $e^x = \sum \frac{x^n}{n!}$ and $e^{x^2} = \sum \frac{x^{2n}}{n!}$. Therefore

$$e^{x+x^2} = \sum_{n=0}^{\infty} \left(\sum_{\substack{0 \leq k \leq n \\ n-k \text{ even}}} \frac{1}{k! \binom{n-k}{2}!} \right) x^n = 1 + x + \frac{3}{2}x^2 + \dots$$

□

Problem 2.4.11. *Approximate $\cos 0.1$ under error 10^{-3} .*

Proof. Recall the problem 1.7.4. Note the power series of $\cos x$ is an alternating series which obviously converge at $x = 0.1$. Then we need to find n such that $\left| \frac{(0.1)^{2n}}{(2n)!} \right| < 10^{-3}$. It is clear that $n = 2$ makes sense. Therefore

$$\left| \cos 0.1 - \left(1 - \frac{(0.1)^2}{2!} \right) \right| \leq 10^{-3}.$$

□

Problem 2.4.3. Show that there exist analytic functions (i.e. functions that can be represented by power series) $u, v : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\sin x = x - \frac{x^3}{6}u(x), \quad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24}v(x)$$

Also, find the radius of convergence and $u(0), v(0)$.

Proof. Observe that $u(x) = 6 \frac{\sin x - x}{x^3} = 6 \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n-2}$ where $x \neq 0$. We have an issue at $x = 0$. But since u is power series, it is continuous. Therefore,

$$u(0) = \lim_{x \rightarrow 0} u(x) = 1.$$

Also This series obviously converges on whole $\mathbb{R}$. Therefore, we get

$$u(x) = 6 \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n-2} \quad x \in \mathbb{R}.$$

Similarly, we can deduce that $v(x) = 24 \sum_{n=2}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n-4}$, and $v(0) = 1$.

In general, we may think it as a partial series, obtained by removing few terms at front and dividing appropriately. □

Problem 2.4.4. Using problem 3, calculate

1. $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

2. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2}$

3. $\lim_{x \rightarrow 0} \frac{\cos x - 1 + x^2/2}{x^4}$

Proof. For (1) and (3), suffices to calculate with u and v . For (2), consider the power series $w(x)$ satisfying $\cos x = 1 - \frac{x^2}{2}w(x)$. Then we can get:

1.

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{x^3 u(x)}{6x^3} = \lim_{x \rightarrow 0} \frac{1}{6} u(x) = \frac{1}{6} u(0) = \frac{1}{6}$$

2.

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{x^2 w(x)}{2x^2} = \lim_{x \rightarrow 0} \frac{1}{2} w(x) = \frac{1}{2} w(0) = \frac{1}{2}$$

3.

$$\lim_{x \rightarrow 0} \frac{\cos x - 1 + x^2/2}{x^4} = \lim_{x \rightarrow 0} \frac{x^4 v(x)}{24x^4} = \lim_{x \rightarrow 0} \frac{1}{24} v(x) = \frac{1}{24} v(0) = \frac{1}{24}$$

□

Problem 2.4.7. *Show that*

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots$$

on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Proof. Since $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $|\cos x - 1| < 1$. Therefore,

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} = \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots}{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots} \\ &= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \left(1 + \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \cdots\right) + \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \cdots\right)^2 + \cdots\right) \\ &= x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots \end{aligned}$$

□

Problem 2.5.2. Find the power series representation of the following functions:

1. $f(x) = \cosh(x^2)$

2. $g(x) = \begin{cases} \frac{\sinh x}{x} & (x \neq 0) \\ 1 & (x = 0) \end{cases}$

3. $h(x) = \frac{\cosh x}{1 - x}$

Proof.

1. Just put x^2 into the power series representation of $\cosh x$. Therefore,

$$\cosh(x^2) = \sum_{n=0}^{\infty} \frac{x^{4n}}{(2n)!} \quad (x \in \mathbb{R})$$

2. As we did in the problem 2.4.3, it is easy to see that $g(x) = \frac{\sinh x}{x} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n+1)!}$ for $x \neq 0$. What we need to check is continuity at $x = 0$.

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n+1)!} = 1,$$

hence $g(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n+1)!}$ for all $x \in \mathbb{R}$.

3. Use the problem 2.3.10. Then we get:

$$h(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(2k)!} \right) x^n \quad (\text{where } \lfloor n/2 \rfloor \text{ denotes the greatest integer } \leq n/2.)$$

□

Problem 2.5.5. For $f(x) = \sum f_n x^n$, show that, for all $n \in \mathbb{N}$,

1.

$$f_{2n+1} = 0$$

$$\text{if } f(x) = f(-x).$$

2.

$$f_{2n} = 0$$

$$\text{if } f(x) = -f(-x).$$

Proof.

1.

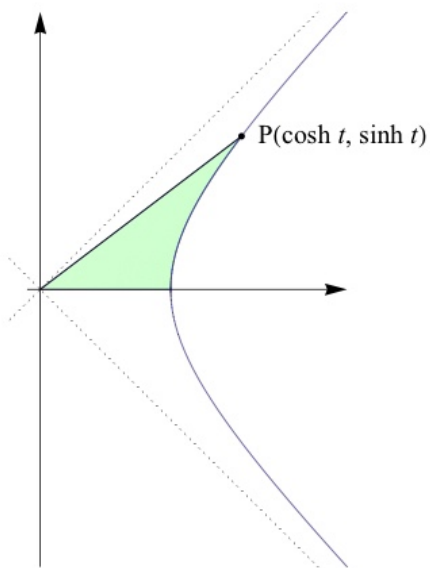
$$\begin{aligned} f(x) - f(-x) &= \sum (1 - (-1)^n) f_n x^n \equiv 0 \\ \Rightarrow f_{2n+1} &= 0 \end{aligned}$$

2.

$$\begin{aligned} f(x) + f(-x) &= \sum (1 + (-1)^n) f_n x^n \equiv 0 \\ \Rightarrow f_{2n} &= 0 \end{aligned}$$

□

Problem 2.5.9. Show that the area enclosed between the line from the origin to $P = (\cosh t, \sinh t)$ and the hyperbola $x^2 - y^2 = 1$ is equal to $\frac{t}{2}$ on $t \geq 0$ and $x, y \geq 0$.



Proof. Put the area function

$$S(t) = \frac{1}{2} \cosh t \sinh t - \int_1^{\cosh t} \sqrt{x^2 - 1} \, dx.$$

Put $x = \cosh u$, then we get

$$\begin{aligned} \int_1^{\cosh t} \sqrt{x^2 - 1} \, dx &= \int_0^t \sinh u \cdot \sinh u \, du \\ &= \frac{1}{2} \int_0^t \cosh 2u - 1 \, du \\ &= \frac{1}{4} \sinh 2t - \frac{t}{2} \\ &= \frac{1}{2} \cosh t \sinh t - \frac{t}{2} \end{aligned}$$

□

Problem 2.6.5. *Show that*

$$\arcsin x = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{2^n n! (2n+1)} x^{2n+1} = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \cdots \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Proof. Since $(\sin x)' = \cos x \neq 0$ if $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, then by the IFT, we get $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$. Of course $|x| < 1$. By the problem 2.3.7, we get

$$\frac{1}{\sqrt{1-x^2}} = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{2^n n!} x^{2n}.$$

Integrate both sides, then we get:

$$\arcsin x = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{2^n n! (2n+1)} x^{2n+1} \quad |x| < 1.$$

□

Problem 2.6.14. Let $f : I_1 \rightarrow I_2$ be a bijective function between open intervals I_1 and I_2 , and suppose that f is twice differentiable with $f'(x) \neq 0$ for all $x \in I_1$. Show that the inverse function $g(x) := f^{-1}(x)$ is also twice differentiable, and

$$g''(y) = -\frac{f''(x)}{(f'(x))^3}$$

for $x \in I_1$ and $y \in I_2$ such that $f(x) = y$

Furthermore, show that if $f(x)$ is three times differentiable, then $g(x)$ is also three times differentiable, and

$$g'''(y) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}$$

More generally, show that if $f(x)$ is n -times differentiable, so is $g(x)$.

Proof. Put $f(x) = y$ for some $x \in I_1$. By the inverse function theorem, the inverse of f exists, which is differentiable. Then we get

$$g'(y) = \frac{1}{f'(g(y))}.$$

Write $h(x) = \frac{1}{x}$. The fact that f' is nowhere zero implies that $g'(x) = h \circ (f' \circ g)(y)$ is differentiable. Therefore,

$$g''(y) = -\frac{f''(g(y))g'(y)}{\{f'(g(y))\}^2} = -\frac{f''(g(y))}{\{f'(g(y))\}^3} = -\frac{f''(x)}{\{f'(x)\}^3}.$$

Since $x \in I_1$ was arbitrarily selected, we are done. To show the second formula, we can follow exactly the same logic.

To show that last statement, we can just use induction. Follow these steps:

1. Assume that f and g are differentiable for $k + 1$ times. Show that if we differentiate $y = f(g(y))$ for k times, the result can be represented by at most k times differentiated f and g , using induction.
2. Show that if f is differentiable k times, then g is differentiable k times by using (1).

□

Problem 2.6.17.

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

Answer the following questions about the function above.

1. Find the value of $f\left(\frac{1}{\sqrt{3}}\right)$.
2. Show that if $|x| < 1$, then f has the inverse function g , and find $g'\left(\frac{\pi}{6}\right)$.

Proof. Observe that

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}.$$

Also, we have shown that $(\arctan x)' = \frac{1}{1+x^2}$. Therefore $f(x) = \arctan x$.

1. $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$. Note that we can put $\frac{1}{\sqrt{3}}$ into $\arctan x$ instead of $f(x)$ because f converges to $\arctan x$ if $|x| < 1$. Be careful to check the radius of convergence!
2. Since $|x| < 1$, we can say $f(x) = \arctan x$. Therefore $g'\left(\frac{\pi}{6}\right) = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$.

□

Problem 2.6.21. Let $y = \tanh x$, and write its inverse by $x = \tanh^{-1} y$. Show that

$$\tanh^{-1} y = \frac{1}{2} \log \frac{1+y}{1-y} = y + \frac{y^3}{3} + \frac{y^5}{5} + \cdots \quad (|y| < 1)$$

Using this, show that

$$\frac{d}{dy} (\tanh^{-1} y) = \frac{1}{1-y^2} \quad (*)$$

Also, show that

$$\frac{d}{dx} \tanh x = 1 - \tanh^2 x$$

and use it to derive (*).

Proof. By the definition, we get:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

From that, we can derive the following:

$$\begin{aligned} y &= \frac{e^{2x} - 1}{e^{2x} + 1} \\ \Leftrightarrow (e^{2x} + 1)y &= e^{2x} - 1 \\ \Leftrightarrow 1 + y &= (1 - y)e^{2x} && \cdots \text{Assume that } |y| < 1. \\ \Leftrightarrow e^{2x} &= \frac{1+y}{1-y} \\ \Leftrightarrow x &= \frac{1}{2} \log \frac{1+y}{1-y} = \tanh^{-1} y. \end{aligned}$$

Also, if $|y| < 1$, by simply differentiating, we get

$$\frac{d}{dy} \tanh^{-1} y = \frac{1}{1-y^2}.$$

Last statement can be obtained by some simple calculations. □

5 5th week

5.1 A Breif Recall of The Last Week

We have observed some approximation of certain functions. But we actually didn't derived some approximations of horrible-shaped functions; we only watched the approximated forms! Someone can ask that "Well, where did those formulas come from?" From now on, we are going to discuss how to approximate some functions by ourselves.

Definition 5.1.1. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, we say that f is $o(x^n)$ if

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^n} = 0.$$

Intuitively, $o(x^n)$ gets smaller as n gets bigger. As we want to **approximate** some function, say f , we want to make $f(x)$ very close to some polynomial, say $p(x)$. To represent f and p are very close, we write $f(x) - p(x) = o(x^n)$. If we can make such $p(x)$ for $f(x)$, we call $p(x)$ by n -th **degree approximation polynomial**. This is why we are defining $o(x^n)$. As you can see, we would be happy to have higher degree approximation polynomial.

The following theorem would be **the most important** theorem in this class, which states how to make approximation polynomials.

Theorem (Taylor's Theorem). Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable for n times at the origin. Then we can always make the unique n -th degree approximation polynomial, which is

$$T_n(f)(x) = f(0) + f'(0)x + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Definition 5.1.2. Write

$$R_n(f)(x) = f(x) - T_n(f)(x).$$

We call $R_n(f)(x)$ by n -th **degree remainder terms** of f . We call the following expression

$$f(x) = T_n(f)(x) + R_n(f)(x)$$

by the **Taylor expansion** at the origin. It is clear that $R_n(f)(x) = o(x^n)$.

We learn L'Hôpital's rule to easily calculate n -th derivative of functions. We first state L'Hôpital's rule.

Theorem. *Suppose that we have one of the following cases,*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \quad \text{OR} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\pm\infty}{\pm\infty}$$

where a can be any real number, infinity or negative infinity. In these cases we have,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

5.2 Problems

Problem 3.2.1. *Evaluate the following limits.*

- | | |
|--|--|
| (1) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ | (2) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ |
| (3) $\lim_{x \rightarrow 0} \frac{x - \tan x}{x - \sin x}$ | (4) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x}$ |
| (5) $\lim_{x \rightarrow 0} \frac{e^x - 1}{\log(1 + x)}$ | (6) $\lim_{x \rightarrow 0} (e^x + x)^{1/x}$ |
| (7) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$ | (8) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ |
| (9) $\lim_{x \searrow 0} \sin x \log x$ | (10) $\lim_{x \searrow 0} \sqrt{x} \log x$ |
| (11) $\lim_{x \searrow 0} x^a \log x \quad (a > 0)$ | (12) $\lim_{x \searrow 0} x^{x^x}$ |
| (13) $\lim_{x \searrow 0} x^{\sin x}$ | (14) $\lim_{x \searrow 0} \frac{\sqrt{x}}{\sqrt{x} + \sin \sqrt{x}}$ |

Proof.

- (1) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}$
- (2) $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$
- (3) $\lim_{x \rightarrow 0} \frac{x - \tan x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{-2 \sec^2 x \tan x}{\sin x}$
- (4) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x} = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{\cos x} = \frac{2}{1} = 2$
- (5) $\lim_{x \rightarrow 0} \frac{e^x - 1}{\log(1+x)} = \lim_{x \rightarrow 0} \frac{e^x}{\frac{1}{1+x}} = 1$
- (6) $\lim_{x \rightarrow 0} (e^x + x)^{1/x} = \exp \left(\lim_{x \rightarrow 0} \frac{\log(e^x + x)}{x} \right) = \exp \left(\lim_{x \rightarrow 0} \frac{e^x + 1}{e^x + x} \right) = e^2$
- (7) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x + x \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{2 \cos x - x \sin x} = 0$
- (8) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \lim_{x \rightarrow 0} \lim_{x \rightarrow 0} \frac{\sec^2 x - \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x + \sin x}{6x}$
 $= \lim_{x \rightarrow 0} \frac{2 \cdot [2 \sec^2 x \tan^2 x + \sec^4 x] + \cos x}{6} = \frac{1}{2}$
- (9) $\lim_{x \rightarrow 0^+} \sin x \log x = \lim_{x \rightarrow 0^+} \frac{\log x}{\csc x} = \lim_{x \rightarrow 0^+} \frac{1}{x(-\csc x \cot x)} = \lim_{x \rightarrow 0^+} \frac{-\sin x \tan x}{x} = 0$
- (10) $\lim_{x \rightarrow 0^+} \sqrt{x} \log x = \lim_{x \rightarrow 0^+} \frac{\log x}{\frac{1}{\sqrt{x}}} = \lim_{x \rightarrow 0^+} -2 \frac{x \sqrt{x}}{x} = 0$
- (11) $\lim_{x \rightarrow 0^+} x^a \log x = \lim_{x \rightarrow 0^+} \frac{\log x}{x^{-a}} = \lim_{x \rightarrow 0^+} \frac{x^{a+1}}{-ax} = 0 \quad (a > 0)$
- (12) $\lim_{x \rightarrow 0^+} x^{x^x} = \exp \left(\lim_{x \rightarrow 0^+} x^x \log x \right) = \exp \left(\lim_{x \rightarrow 0^+} \log x \right) = 0.$
 $\therefore \lim_{x \rightarrow 0^+} x^x = 1.$
- (13) $\lim_{x \rightarrow 0^+} x^{\sin x} = \exp \left(\lim_{x \rightarrow 0^+} \sin x \log x \right) = \exp(0) = 1$
- (14) $\lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sqrt{x} + \sin \sqrt{x}} = \lim_{x \rightarrow 0} \frac{1}{1 + \frac{\sin \sqrt{x}}{\sqrt{x}}} = \frac{1}{2}$

□

Problem 3.2.6. *Show the followings:*

$$\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0, \quad \lim_{x \rightarrow \infty} \frac{\log x}{\sqrt{x}} = 0.$$

In general, for any natural number n , show the following:

$$\lim_{x \rightarrow \infty} \frac{\log x}{\sqrt[n]{x}} = 0.$$

Proof. We just show the last statement. This is direct from the L'Hôpital's rule:

$$\lim_{x \rightarrow \infty} \frac{\log x}{x^{\frac{1}{n}}} = \lim_{x \rightarrow \infty} n \frac{x^{\frac{n-1}{n}}}{x} = 0$$

□

Problem 3.2.7. For positive real numbers a, b, c , show the following:

1. $\lim_{n \rightarrow \infty} \left(\frac{a^{1/n} + b^{1/n}}{2} \right)^n = \sqrt{ab}$
2. $\lim_{n \rightarrow \infty} \left(\frac{a^{1/n} + b^{1/n} + c^{1/n}}{3} \right)^n = \sqrt[3]{abc}$

Proof.

1. Suppose either a or b is 0, then the result is trivial. Therefore WMA $a, b > 0$. Put $x = \frac{1}{n}$, and take logarithm, then we get:

$$\frac{\log(a^x + b^x) - \log 2}{x}$$

Use L'Hôpital's Rule:

$$\lim_{x \searrow 0} \frac{a^x \log a + b^x \log b}{a^x + b^x} = \log \sqrt{ab}$$

Therefore, the limit of the original expression is $\sqrt{ab}$.

2. If $c = 0$, then:

$$\lim_{n \rightarrow \infty} \left(\frac{a^{1/n} + b^{1/n}}{3} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{a^{1/n} + b^{1/n}}{2} \right)^n \cdot \left(\frac{2}{3} \right)^n = 0$$

The result also vanishes when a or b is 0. So WMA $a, b, c > 0$. Taking logarithms and setting $x = \frac{1}{n}$, we get:

$$\lim_{x \searrow 0} \frac{\log(a^x + b^x + c^x) - \log 3}{x} = \lim_{x \searrow 0} \frac{a^x \log a + b^x \log b + c^x \log c}{a^x + b^x + c^x} = \log \sqrt[3]{abc}$$

Therefore, the limit of the given expression is $\sqrt[3]{abc}$.

□

Problem 3.3.4. Find the third-degree Taylor polynomial of the following functions at the origin.

$$\begin{array}{ll}
 (1) f(x) = e^x \sin x & (2) f(x) = x \log(1+x) \\
 (3) f(x) = x - \cos x & (4) f(x) = 2^x \\
 (5) f(x) = \sqrt{1+x} & (6) f(x) = (1+x)^{-1} \\
 (7) f(x) = \frac{1}{(x+1)(x+2)} & (8) f(x) = \frac{1}{x^2+2x+2}
 \end{array}$$

Proof. I will omit the proof if it is a tedious calculation.

$$1. T_3(f)(x) = x + x^2 + \frac{1}{3}x^3.$$

We can just multiply the series expression of e^x and $\sin x$, rather than differentiating.

$$2. T_3(f)(x) = -1 + x + \frac{1}{2}x^2.$$

$$3. T_3(f)(x) = -1 + x + \frac{1}{2}x^2.$$

$$4. T_3(f)(x) = 1 + x \log 2 + \frac{1}{2}x^2(\log 2)^2 + \frac{1}{6}x^3(\log 2)^3.$$

$$5. T_3(f)(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3.$$

Use **binomial expansion**, i.e. problem 2.3.7.

$$6. T_3(f)(x) = 1 - x + x^2 - x^3$$

$$7. T_3(f)(x) = \frac{1}{2} - \frac{3}{4}x + \frac{7}{8}x^2 - \frac{15}{16}x^3.$$

Use the fact that $\frac{1}{2+x} = \frac{1}{2} \frac{1}{1+\frac{x}{2}}$. Multiply two series expansions.

$$8. T_3(f)(x) = \frac{1}{2} - \frac{1}{2}x + \frac{1}{4}x^2.$$

Use the fact that $\frac{1}{2+2x+x^2} = \frac{1}{2} \frac{1}{1+(x+\frac{x^2}{2})}$.

□

Problem 3.3.6. Suppose that a twice differentiable function at the origin $f : \mathbb{R} \rightarrow \mathbb{R}$ is given. If $T_2(f)(x) = 1 + 2x + 4x^2$, then what is $T_2\left(\frac{1}{f}\right)(x)$?

Proof. First note that $\frac{1}{f}$ is well defined near the origin since $f(0) = 1$. Clearly $\frac{1}{f}$ is differentiable. With some calculation, we get:

$$\begin{aligned}\left(\frac{1}{f}\right)'(0) &= -\frac{f'(0)}{f(0)^2} = -2 \\ \left(\frac{1}{f}\right)''(0) &= \frac{2f'(0)^2 - f(0)f''(0)}{f(0)^3} = 0\end{aligned}$$

Therefore $T_2\left(\frac{1}{f}\right)(x) = 1 - 2x$. □

Problem 3.3.8. Show that if a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is $o(x^n)$, then $F(x) := \int_0^x f(t)dt$ is $o(x^{n+1})$.

Proof.

$$\lim_{x \rightarrow 0} \frac{F(x)}{x^{n+1}} = \lim_{x \rightarrow 0} \frac{f(x)}{(n+1)x^n} = 0.$$

□

Problem 3.3.14. Define

$$f(x) = \begin{cases} x^3 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Show that

1. $f'(0) = 0$
2. $f(x)$ is differentiable.
3. $f(x) = o(x^2)$.
4. $f''(0)$ is not defined.

Why $f(x)$ and the theorem 2.3.1 is compatible?

Theorem (2.3.1). Suppose that f is differentiable n -times. Then $f(x) = o(x^n)$ if and only if

$$f(0) = 0, f'(0) = 0, \dots, f^{(n)}(0) = 0.$$

□

Proof. (1)-(3) are tedious calculation. We get

$$f'(x) = \begin{cases} 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x} & x \neq 0 \\ 0 & x = 0. \end{cases}$$

We can observe that

$$\lim_{x \rightarrow 0} \frac{f'(x)}{x}$$

does not exist. It is compatible with the theorem because it is not twice differentiable.

□

6 6th week

6.1 A Brief Recall of The Last Week

Last week, we have discussed about **Taylor Series**. But, actually, how can we guarantee that the series we calculated indeed approximates the original function? Even though I **said** that we can get better approximation whenever n gets bigger, someone can think that I am lying. How can I prove that I am not? That is, how can I say that n -th degree remainder term converges to 0 as n gets bigger?

6.2 Lagrange's Remainder Theorem

The Lagrange's Remainder theorem actually guarantees you to **believe** me. But before that, in order to prove the truth-telling theorem, we need one lemma:

Lemma (Cauchy's Mean Value Theorem). *For functions f, g which are continuous on $[a, b]$ and differentiable on (a, b) , $\exists c \in (a, b)$ such that*

$$\frac{f(b) - f(a)}{b - a} : \frac{g(b) - g(a)}{b - a} = f'(c) : g'(c)$$

Proof. See page 143 in our text book, Lemma 5.1.1. □

Theorem (Lagrange's Remainder term). *Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable for $n + 1$ times. Then, for some $0 < \xi < x$, the following holds:*

$$f(x) = \sum_{j=0}^n \frac{f^{(j)}(0)}{j!} x^j + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} x^{n+1}}_{=R_n(f)(x)}$$

Proof. Define

$$F(\xi) = f(\xi) + \sum_{k=1}^n \frac{f^{(k)}(\xi)}{k!} (x - \xi)^k.$$

Observe that $F(x) = f(x)$, and $F(0) = T_n(f)(x)$. Then we get $f(x) - T_n(f)(x) = F(x) - F(0)$. Since LHS is differentiable, so is RHS. Hence we get:

$$\begin{aligned} F'(\xi) &= f'(\xi) + \sum_{k=1}^n \left[\frac{f^{(k+1)}(\xi)}{k!} (x - \xi)^k - \frac{f^{(k)}(\xi)}{(k-1)!} (x - \xi)^{k-1} \right] \\ &= \frac{f^{(n+1)}(\xi)}{n!} (x - \xi)^n. \end{aligned}$$

We first assume that $x > 0$. Observe that $G(\xi) = (x - \xi)^{n+1}$ is continuous on $[0, x]$ and differentiable over

$(0, x)$. Now we apply the CMVT:

$$\begin{aligned}
 F(x) - F(0) &= \frac{F'(\xi)}{G'(\xi)}[G(x) - G(0)] \quad \text{for some } \xi \in (0, x) \\
 \Rightarrow f(x) - T_n(f)(x) &= \frac{\frac{f^{(n+1)}(\xi)}{n!}(x - \xi)^n}{-(n+1)(x - \xi)^n}[0 - x^{n+1}] = \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1} \quad \text{for some } \xi \in (0, x) \\
 \therefore R_n(f)(x) &= \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1} \quad \text{for some } \xi \in (0, x).
 \end{aligned}$$

The case where $x < 0$ follows exactly same logic. □

Corollary.

$$|R_n(f)(x)| \leq \frac{\max \{|f^{(n+1)}(\xi)| : \xi \in [0, x]\}}{(n+1)!}|x|^{n+1}$$

□

Also, we can calculate taylor series even at $x = a$ where a is arbitrary. This is simple; just put $x - a$ instead of x . The same logic works.

6.3 Problems

Problem 3.4.5. *In the n th-degree Taylor expansion of e^x , show that $R_n(x)$ satisfies*

$$|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \max\{e^x, 1\},$$

and approximate e under error 0.001.

Proof. This is direct from the corollary in the previous page. Put $x = 1$. Then we get

$$|R_6(1)| \leq \frac{e}{7!} < \frac{3}{5040} \leq 0.001.$$

Therefore

$$e \approx 2 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!}.$$

□

Problem 3.4.7. Calculate followings under error 10^{-3} .

1. $\int_0^{0.3} e^{-x^2} dx$

2. $\int_0^1 \frac{\sinh x}{x} dx$

3. $\int_0^1 \frac{\sin x}{x} dx$

Proof.

1.

$$\int_0^{0.3} e^{-x^2} dx = \sum_{n=0}^{\infty} \frac{(-1)^n (0.3)^{2n+1}}{(2n+1) \cdot n!}.$$

is trivial. By the problem 1.7.4, we get:

$$\left| \int_0^{0.3} e^{-x^2} dx - \sum_{n=0}^1 \frac{(-1)^n (0.3)^{2n+1}}{(2n+1) \cdot n!} \right| \leq \frac{(0.3)^5}{5 \cdot 2!} \leq 10^{-3}.$$

2. By the Taylor expansion of $\sinh x$, we get:

$$\int_0^1 \frac{\sinh x}{x} dx = \sum_{n=0}^{\infty} \frac{1}{(2n+1)(2n+1)!}.$$

Observe that:

$$\sum_{n=k+1}^{\infty} \frac{1}{(2n+1)(2n+1)!} \leq \sum_{n=k+1}^{\infty} \frac{1}{(2k+3)(2n+1)!} = \frac{\sum_{n=k+1}^{\infty} \frac{(2k+3)!}{(2n+1)!}}{(2k+3)(2k+3)!} < \frac{e}{(2k+3)(2k+3)!}$$

Put $k = 2$, then $\frac{e}{7 \cdot 7!} \leq 10^{-3}$, hence we get:

$$\left| \int_0^1 \frac{\sinh x}{x} dx - \left(1 + \frac{1}{3 \cdot 3!} + \frac{1}{5 \cdot 5!} \right) \right| \leq 10^{-3}.$$

3. By the same logic with (1),

$$\left| \int_0^1 \frac{\sin x}{x} dx - \left(1 - \frac{1}{3 \cdot 3!} + \frac{1}{5 \cdot 5!} \right) \right| \leq \left| -\frac{1}{7 \cdot 7!} \right| \leq 10^{-3}.$$

□

Problem 3.4.13. Let $g(x) = \frac{x}{1+3x^2}$. Where does $T(g)(x)$ converge to $g(x)$?

Proof. By the expression of $\frac{1}{1+x}$ on $|x| < 1$, we get:

$$\frac{x}{1+3x^2} = \sum_{n=1}^{\infty} (-3)^n x^{2n+1} \quad |x| < \frac{1}{\sqrt{3}}.$$

□

Problem 3.4.16. Let $f(x)$ be an n -times differentiable function defined on an interval containing the origin. Show that there exists x_* between 0 and x such that

$$R_n(x) = \frac{f^{(n)}(x_*) - f^{(n)}(0)}{n!} x^n.$$

Moreover, define

$$B_n(x) := \max \left\{ \left| f^{(n)}(t) - f^{(n)}(0) \right| : t \in [0, x] \right\},$$

and show that the following inequality holds:

$$|R_n(x)| \leq B_n(x) \frac{|x|^n}{n!}.$$

Proof. As we showed, $R_{n-1}(f)(x) = \frac{f^{(n)}(\xi)}{n!} x^n$ for some $\xi \in (0, x)$. On the other hand, by the definition, we know that $R_n(f)(x) = R_{n-1}(f)(x) - \frac{f^{(n)}(0)}{n!} x^n$. Therefore

$$\begin{aligned} R_n(f)(x) &= R_{n-1}(f)(x) - \frac{f^{(n)}(0)}{n!} x^n \\ &= \frac{f^{(n)}(\xi) - f^{(n)}(0)}{n!} x^n. \end{aligned}$$

The following inequality is direct from this. □

Problem 3.5.2. Find the n -th degree Taylor polynomial of the following functions at $x = a$.

1. $\sin x, \quad n = 4, \quad a = \frac{\pi}{2}$

2. $\sqrt{1+x}, \quad n = 3, \quad a = 1$

3. $\frac{1}{1+x}, \quad n = 4, \quad a = 2$

4. $\log x, \quad n = 6, \quad a = 1$

Proof.

1. $T_4(x) = 1 - \frac{1}{2} \left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24} \left(x - \frac{\pi}{2}\right)^4$

2. $T_3(x) = \sqrt{2} + \frac{1}{2\sqrt{2}}(x-1) - \frac{1}{16\sqrt{2}}(x-1)^2 + \frac{1}{64\sqrt{2}}(x-1)^3$

3. $T_4(x) = \frac{1}{3} - \frac{1}{9}(x-2) + \frac{1}{27}(x-2)^2 - \frac{1}{81}(x-2)^3 + \frac{1}{243}(x-2)^4$

4. $T_6(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5 - \frac{1}{6}(x-1)^6$

□

Problem 3.5.4. Find the n -th degree Taylor polynomial of the function $f(x) = e^{x^2+2x}$ at $x = -1$.

Proof.

$$f(x) = e^{-1}e^{x^2+2x+1} = \frac{1}{e} \sum_{n=0}^{\infty} \frac{(x+1)^{2n}}{n!} \quad (x \in \mathbb{R})$$

Since it converges everywhere on $\mathbb{R}$, hence we get:

$$T_n f(x) = \frac{1}{e} \sum_{0 \leq 2k \leq n} \frac{(x+1)^{2k}}{k!}$$

□

Problem 3.5.6. For the following functions, find the Taylor series at $x = 3$.

1. $f(x) = x^4$

2. $f(x) = \frac{1}{x}$

Proof.

1. For $f(x) = x^4$, we have:

$$f'(x) = 4x^3, \quad f''(x) = 12x^2, \quad f^{(3)}(x) = 24x, \quad f^{(4)}(x) = 24, \quad f^{(k)}(x) = 0 \quad (k \geq 5)$$

Therefore:

$$\begin{aligned} T(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!} (x-3)^n \\ &= 81 + 108(x-3) + 54(x-3)^2 + 12(x-3)^3 + (x-3)^4 \end{aligned}$$

2. For $f(x) = \frac{1}{x}$, we have:

$$f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}$$

Therefore:

$$T(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!} (x-3)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} (x-3)^n$$

□

7 7th week

From this chapter, we will begin to discuss about “space”. Maybe you are familiar with xyz -coordinate system. But the coordinate system is, in fact, just depend on our “choice”. We can change our coordinate system arbitrarily! In this chapter, we are going to use the following coordinate system:

7.1 Polar Coordinate System

$$(x, y) \mapsto (r \cos \theta, r \sin \theta)$$

7.2 Cylindrical Coordinate System

$$(x, y, z) \mapsto (r \cos \theta, r \sin \theta, z)$$

It is just an extension of polar coordinate system into 3-dimensional space.

7.3 Spherical Coordinate System

$$(x, y, z) \mapsto (\rho \sin \varphi \cos \theta, \rho \sin \varphi \sin \theta, \rho \cos \varphi)$$

Here, $\rho \geq 0$, $0 \leq \varphi \leq \pi$, and $0 \leq \theta < 2\pi$.

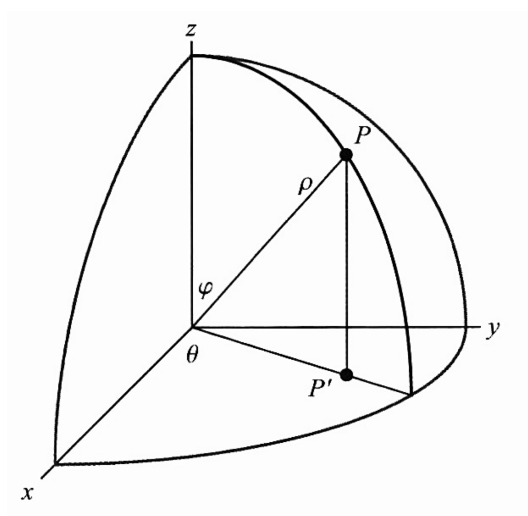


Figure 1: Spherical Coordinate System

7.4 Conics in Polar Coordinate System

The generalized polar equation of conics is given by:

$$r = \frac{a}{1 + e \cos \theta} \quad (e \geq 0)$$

- Circle: $e = 0$
- Ellipse: $0 < e < 1$.
- Parabola: $e = 1$.
- Hyperbola: $e > 1$.

7.5 Problems

Problem 4.1.5. In 3-dimensional Cartesian space, let a line segment of length ℓ have projections onto the yz -, zx -, and xy -planes with lengths ℓ_1, ℓ_2, ℓ_3 , respectively.

Show that

$$\ell_1^2 + \ell_2^2 + \ell_3^2 = 2\ell^2,$$

and

$$|\ell_1^2 - \ell_2^2| \leq \ell_3^2 \leq \ell_1^2 + \ell_2^2.$$

Proof. Let the given line segment be $L = \overline{AB}$. WLOG, WMA $A = O$. Write $B = (x, y, z)$. Then clearly $\ell = \sqrt{x^2 + y^2 + z^2}$, and

$$\ell_1^2 = y^2 + z^2$$

$$\ell_2^2 = x^2 + z^2$$

$$\ell_3^2 = x^2 + y^2$$

Therefore the first equality holds. Also,

$$|\ell_1^2 - \ell_2^2| = |y^2 - x^2| \leq x^2 + y^2 = \ell_3^2 \leq x^2 + y^2 + 2z^2 = \ell_1^2 + \ell_2^2.$$

□

Problem 4.1.9. Consider the set of all polynomials of degree less than or equal to 3:

$$\{a_0 + a_1x + a_2x^2 + a_3x^3 \mid a_0, a_1, a_2, a_3 \in \mathbb{R}\}.$$

Describe a one-to-one correspondence between this set and the 4-dimensional coordinate space $\mathbb{R}^4$. Does the sum of two polynomials correspond to the sum of two points in the space?

Proof. Use the map

$$a_0 + a_1x + a_2x^2 + a_3x^3 \mapsto (a_0, a_1, a_2, a_3).$$

The remainings are clear. □

Problem 4.2.4. Convert the following polar equations of curves into Cartesian equations and sketch their shapes.

(1) $r = \sin \theta$

(2) $r = 2 \cos \theta$

(3) $r = \sin 3 \left(\theta + \frac{\pi}{2} \right)$

Proof.

(1) $r^2 = r \sin \theta \Rightarrow x^2 + y^2 = y$

(2) $r^2 = 2r \cos \theta \Rightarrow x^2 + y^2 = 2x$

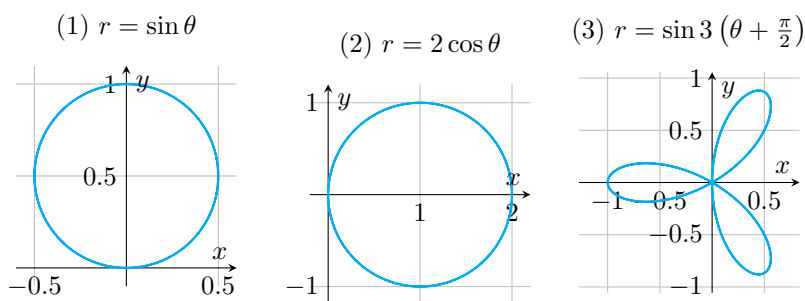
(3)

$$\begin{aligned} r^4 &= r^3 \sin 3 \left(\theta + \frac{\pi}{2} \right) = -r^3 \cos(3\theta) = 3r^3 \cos \theta - 4r^3 \cos^3 \theta \\ &\Rightarrow (x^2 + y^2)^2 = 3(x^2 + y^2)x - 4x^3 \end{aligned}$$

Note.

$$\sin(3\theta) = 3 \sin \theta - 4 \sin^3 \theta$$

$$\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta$$



□

Problem 4.2.6. Sketch the shape of the curves given below in polar coordinates.

(1) $r = |\cos 2\theta|$

(3) $r^2 = \sin \theta$

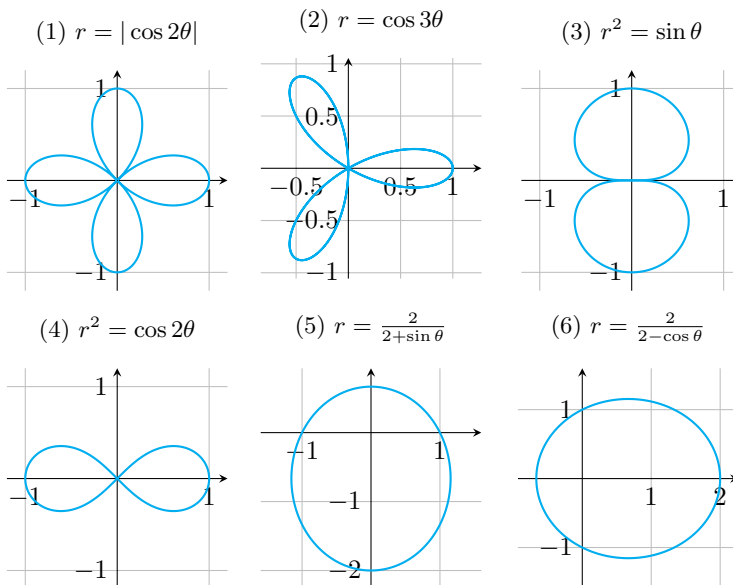
(5) $r = \frac{2}{2 + \sin \theta}$

(2) $r = \cos 3\theta$

(4) $r^2 = \cos 2\theta$

(6) $r = \frac{2}{2 - \cos \theta}$

Proof.



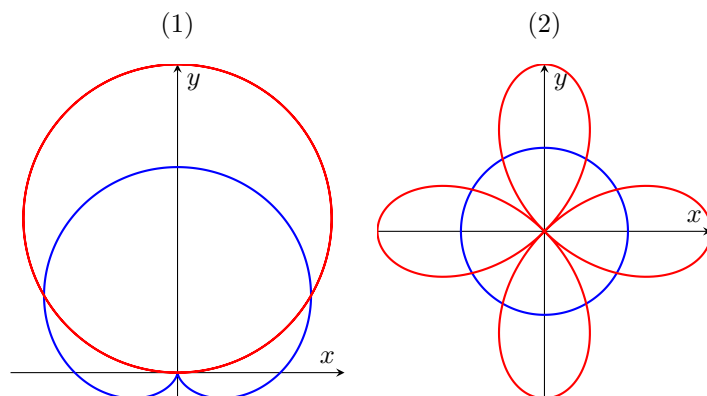
□

Problem 4.2.7. Sketch the region in the xy -plane that satisfies both of the following polar inequalities.

1. $r \leq 3 \sin \theta, \quad r \geq 1 + \sin \theta$

2. $r \leq \cos 2\theta, \quad r \geq \frac{1}{2}$

Proof. Plot the curves:



□

Problem 4.2.9. 1. On the right half-plane $x > 0$, let

$$\theta(x, y) := \arctan \frac{y}{x}$$

Show that the following equations are satisfied:

$$x = r \cos \theta(x, y), \quad y = r \sin \theta(x, y) \quad (*)$$

2. On the left half-plane $x < 0$, let

$$\theta(x, y) := \pi + \arctan \frac{y}{x}$$

Show that equation $(*)$ holds.

3. On the upper half-plane $y > 0$, let

$$\theta(x, y) := \frac{\pi}{2} + \arctan \left(\frac{-x}{y} \right) = \frac{\pi}{2} - \arctan \left(\frac{x}{y} \right)$$

Show that equation $(*)$ holds.

4. On the lower half-plane $y < 0$, let

$$\theta(x, y) := -\frac{\pi}{2} + \arctan \left(\frac{x}{-y} \right) = -\frac{\pi}{2} - \arctan \left(\frac{x}{y} \right)$$

Show that equation $(*)$ holds.

5. If some function $\theta(x, y)$ satisfies equation $(*)$, show that for any integer n , the function

$$\tilde{\theta}(x, y) := \theta(x, y) + 2n\pi$$

also satisfies equation $(*)$.

Proof. Suffice to show that $\theta(x, y)$ equals to the degree of (x, y) with respect to x -axis. It is clear if we apply some appropriate rotation. Also, (5) is clear by the period of $\sin$ and $\cos$. $\square$

Problem 4.3.1. Express the following points in rectangular coordinates as cylindrical coordinates and spherical coordinates.

1. $(1, \sqrt{3}, 0)$

2. $(-1, -1, -1)$

Proof.

(1) Cylindrical coordinates:

$$(r, \theta, z) = \left(2, \frac{\pi}{3}, 0\right)$$

Spherical coordinates:

$$(\rho, \varphi, \theta) = \left(2, \frac{\pi}{2}, \frac{\pi}{3}\right)$$

(2) Cylindrical coordinates:

$$(r, \theta, z) = \left(\sqrt{2}, \frac{5\pi}{4}, -1\right)$$

Spherical coordinates:

$$(\rho, \varphi, \theta) = \left(\sqrt{3}, \arccos\left(-\frac{1}{\sqrt{3}}\right), \frac{5\pi}{4}\right)$$

□

Problem 4.3.2. Find the distance between the following pairs of points given in spherical coordinates.

1. $(1, \frac{\pi}{6}, \frac{\pi}{4}), (1, \frac{\pi}{3}, \frac{\pi}{4})$
2. $(10, \frac{\pi}{6}, \frac{\pi}{4}), (10, \frac{\pi}{6}, \frac{\pi}{5})$
3. $(5, \frac{\pi}{6}, \frac{\pi}{4}), (5, \frac{\pi}{4}, \frac{\pi}{2})$
4. $(1, \frac{\pi}{2}, \frac{\pi}{3}), (2, \frac{\pi}{6}, \frac{\pi}{3})$

Proof. 1. Since their θ are equal, we can easily compute the distance, just by thinking a triangle.

$$\therefore d = 2\rho \sin\left(\frac{\Delta\varphi}{2}\right) = 2 \cdot 1 \cdot \sin\left(\frac{\pi}{12}\right)$$

2. Since their φ are equal, by the same logic:

$$\therefore d = 2\rho \sin\varphi \sin\left(\frac{\Delta\theta}{2}\right) = 2 \cdot 10 \cdot \sin\left(\frac{\pi}{4}\right) \cdot \sin\left(\frac{\pi}{40}\right) = 10 \sin\left(\frac{\pi}{40}\right)$$

3. Convert both points to rectangular coordinates:

$$\left(5 \sin \frac{\pi}{6} \cos \frac{\pi}{4}, 5 \sin \frac{\pi}{6} \sin \frac{\pi}{4}, 5 \cos \frac{\pi}{6}\right) = \left(\frac{5}{2\sqrt{2}}, \frac{5}{2\sqrt{2}}, \frac{5\sqrt{3}}{2}\right)$$

$$\left(5 \sin \frac{\pi}{4} \cos \frac{\pi}{2}, 5 \sin \frac{\pi}{4} \sin \frac{\pi}{2}, 5 \cos \frac{\pi}{4}\right) = \left(0, \frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$$

Now it's time to enjoy horrible calculation:

$$\sqrt{\left(\frac{5}{2\sqrt{2}}\right)^2 + \left(\frac{5}{2\sqrt{2}} - \frac{5}{\sqrt{2}}\right)^2 + \left(\frac{5\sqrt{3}}{2} - \frac{5}{\sqrt{2}}\right)^2} = 5\sqrt{\frac{3 - \sqrt{6}}{2}}$$

4. Similar to (3), the distance is: $\sqrt{3}$.

□

Problem 4.3.3. Express the following equations given in rectangular coordinates in cylindrical and spherical coordinates.

1. $x^2 + y^2 = z^2$

2. $1 \leq x^2 + y^2 + z^2 \leq 4$

Proof. Just convert coordinate system by the definition.

- (1) – Cylindrical coordinates: $|z| = |r|$
 – Spherical coordinates: $\varphi = \frac{\pi}{4}$ or $\frac{3\pi}{4}$
- (2) – Cylindrical coordinates: $1 \leq r^2 + z^2 \leq 4$
 – Spherical coordinates: $1 \leq \rho \leq 2$

□

Problem 4.3.4. *Convert the following spherical equations into xyz-coordinate equations.*

1. $\rho^2(\sin^2 \theta \sin^2 \varphi + \cos^2 \varphi) = 4$

2. $\rho = \cos \theta \sin \varphi$

Proof. Just convert them again.

1. $y^2 + z^2 = 4$

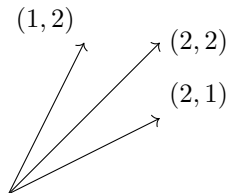
2. $x^2 + y^2 + z^2 = x$

□

8 8th week

8.1 Vector

Maybe you have learned about vector in high school. It's simple, Vector is just an arrow, which has a direction, and a length. Also, if you think about $\mathbb{R}^2$ plane, we can represent a vector as a point in the plane.



Therefore, we can write any point $\vec{v} = (a_1, a_2, \dots, a_n)$ as a vector pointing that point starting from the origin. Note that the vectors which can be identified by a translation are viewed as a same vector. Then we can naturally think about the size of the vector. It is same with the distance of the points!

$$|\vec{v}| = \sqrt{a_1^2 + \dots + a_n^2}.$$

As we can define addition and scalar multiplication of the points, it is same with vectors, just work coordinatewise.

Definition 8.1.1. Let $\vec{u} = (u_1, u_2, \dots, u_n)$ and $\vec{v} = (v_1, v_2, \dots, v_n)$ be vectors in $\mathbb{R}^n$. Then we define

1. Addition: $\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$
2. Scalar multiplication: $c\vec{u} = (cu_1, cu_2, \dots, cu_n)$ for $c \in \mathbb{R}$

We can define the distance of vectors as $|\vec{u} - \vec{v}| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$.

8.2 Dot product

We have one more special operation between vectors, which is not defined between points. This is called **dot product**.

Definition 8.2.1. Let $\vec{u} = (u_1, u_2, \dots, u_n)$ and $\vec{v} = (v_1, v_2, \dots, v_n)$ be vectors in $\mathbb{R}^n$. Then we define the dot product as

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n$$

There are some useful properties of dot product.

Theorem 8.2.2. Let $\vec{u}, \vec{v}, \vec{w}$ be vectors in $\mathbb{R}^n$ and $c \in \mathbb{R}$. Then

1. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
2. $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$

3. $c(\vec{u} \cdot \vec{v}) = (c\vec{u}) \cdot \vec{v} = \vec{u} \cdot (c\vec{v})$
4. $\vec{v} \cdot \vec{v} = |\vec{v}|^2$
5. $\vec{u} \cdot \vec{v} = \frac{1}{2}(|\vec{u} + \vec{v}|^2 - |\vec{u}|^2 - |\vec{v}|^2).$

□

Proofs are easy, so I will omit them. Also, we have another easy theorem:

Theorem 8.2.3. *Let $\vec{u}, \vec{v}$ be vectors in $\mathbb{R}^n$. Then*

1. *For all n -vector $\mathbf{x}$, $\vec{u} \cdot \mathbf{x} = 0$ implies $\vec{u} = 0$.*
2. *For all n -vector $\mathbf{x}$, $\vec{u} \cdot \mathbf{x} = \vec{v} \cdot \mathbf{x}$ implies $\vec{u} = \vec{v}$.*

□

Also, this operation detects the the component of a vector parallel to another.

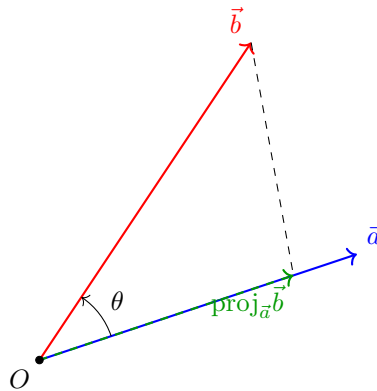
Theorem 8.2.4. *For any vectors $\vec{a}, \vec{b}$ in $\mathbb{R}^n$ which are non-zero, the **projection** of $\vec{b}$ onto $\vec{a}$ is given by*

$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{\vec{a} \cdot \vec{a}} \vec{a}.$$

Also, the angle θ between $\vec{a}$ and $\vec{b}$ is given by

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}.$$

□



8.3 What is a linear transformation?

From now on, we are going to discuss about something related to **vectors**. There is a concept that is deeply related to vector transformation.

Definition 8.3.1. A linear transformation is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u} \in \mathbb{R}^n$ and $c \in \mathbb{R}$

You may think this like a linear function, e.g. $y = ax$. The resulting vector will be on the position of y , and the input vector would be on the position of x . But you may wonder how can we write this transformation as $y = ax$. Well, in fact, to do that, we need to discuss about **matrix**. We will talk about it maybe on next week. For now, let's learn something more about vector.

8.4 Problems

Problem 5.1.1. *Choose all correct statements below.*

1. *A translation is a bijective map.*
2. *In a translation, once you know where one point goes, you know where every point goes.*
3. *For any two points A, B , there exists a unique translation that maps A to B .*
4. *If a map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a translation, then $X \mapsto \text{dist}(X, T(X))$ is a constant function.*

Proof. All of them are true. Note that every translation can be written as $T_w(v) = v + w$. Then we can clearly observe that its inverse is T_{-w} , hence bijective. 2,3 is direct from this fact. Also, since $|x - T_w(x)| = |w|$, it is constant. □

Problem 5.1.2. In n -space, the central symmetry about point P is defined by:

$$S_P(X) = 2P - X \quad (X \in \mathbb{R}^n)$$

This satisfies:

$$\frac{1}{2}(X + S_P(X)) = P$$

Given two points P_1, P_2 , describe the transformation $S_{P_2} \circ S_{P_1}$.

Proof. By the definition,

$$\begin{aligned} S_{P_2} \circ S_{P_1}(X) &= S_{P_2}(2P_1 - X) \\ &= 2P_2 - (2P_1 - X) \\ &= X + 2(P_2 - P_1) \end{aligned}$$

Therefore it's a translation. □

Problem 5.2.1. Given $A = (a_1, \dots, a_n), B = (b_1, \dots, b_n) \in \mathbb{R}^n$:

1. Find the point P such that $\overrightarrow{OP} = \overrightarrow{AB}$, and find the translation T such that $T(\overrightarrow{OP}) = \overrightarrow{AB}$.
2. Given $C = (c_1, \dots, c_n)$, find D such that $\overrightarrow{CD} = \overrightarrow{AB}$, and also find $T(\overrightarrow{CD})$.

Proof.

1.

$$\overrightarrow{OP} = P = (b_1 - a_1, b_2 - a_2, \dots, b_n - a_n) = \overrightarrow{AB}$$

This must be equal to $\overrightarrow{AB}$.

Since T is the translation sending point O to point A ,

$$T(x_1, x_2, \dots, x_n) = (x_1 + a_1, x_2 + a_2, \dots, x_n + a_n)$$

2. Similarly from (1),

$$D = (c_1 + b_1 - a_1, c_2 + b_2 - a_2, \dots, c_n + b_n - a_n)$$

$$T(\overrightarrow{CD}) = B$$

□

Problem 5.2.2. For vectors a, b , compute $a + b$, $a - b$, $2a - 3b$, and $3a + 2b$.

- $a = (2, -1), b = (1, 2)$
- $a = (-1, 2, 2), b = (1, 1, -2)$

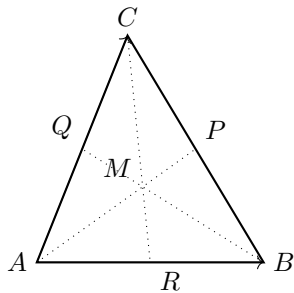
Problem 5.2.3. Find all unit vectors with the same direction as the following vectors:

- $a = (-1, -2, -3)$
- $b = (4, -5, 6)$
- $c = (2021, 2021, 2021)$
- $d = (3, 4, 0)$

Proof. Too easy, omit.

□

Problem 5.2.6. Let M be the centroid of triangle ABC , and let P, Q, R be the midpoints of segments BC, CA, AB respectively. Express the vectors $\overrightarrow{AR}, \overrightarrow{BR}, \overrightarrow{CQ}, \overrightarrow{BP}, \overrightarrow{AM}, \overrightarrow{PM}, \overrightarrow{CM}, \overrightarrow{QM}$ in terms of $s\overrightarrow{AB} + t\overrightarrow{AC}$ for some scalars s, t .



Proof.

$$\overrightarrow{AR} = \frac{1}{2}\overrightarrow{AB},$$

$$\overrightarrow{BR} = -\frac{1}{2}\overrightarrow{AB},$$

$$\overrightarrow{CQ} = -\frac{1}{2}\overrightarrow{AC},$$

$$\overrightarrow{BP} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB},$$

$$\overrightarrow{AM} = \frac{1}{3}\overrightarrow{AB} + \frac{1}{3}\overrightarrow{AC},$$

$$\overrightarrow{PM} = -\frac{1}{6}\overrightarrow{AB} - \frac{1}{6}\overrightarrow{AC},$$

$$\overrightarrow{CM} = \frac{2}{3}\overrightarrow{AC} + \frac{1}{3}\overrightarrow{AB},$$

$$\overrightarrow{QM} = \frac{1}{3}\overrightarrow{AB} - \frac{1}{6}\overrightarrow{AC}.$$

□

Problem 5.3.1. Compute the $\cos$ of the angle between each pair of vectors:

1. $(2, -1), (-1, 1)$

2. $(1, 1), (1, \sqrt{3})$

3. $(1, 2, 3), (3, 2, 1)$

4. $(-1, -2, 3), (4, 5, 6)$

5. $(1, 1, 1, 1), (1, 2, 3, 4)$

6. $(0, 1, 2, 3, 4), (0, 0, 0, 0, 1)$

Proof. Use dot product.

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

1. $\frac{-3}{\sqrt{10}}$

2. $\frac{1 + \sqrt{3}}{2\sqrt{2}}$

3. $\frac{5}{7}$

4. $\frac{7}{\sqrt{22}}$

5. $\frac{5}{\sqrt{30}}$

6. $\frac{4}{\sqrt{30}}$

□

Problem 5.3.3. Compute the projection of vector b onto vector a :

1. $a = (1, -1, 1), b = (2, 1, 5)$

2. $a = (1, -1, 1), b = (2, 3, 1)$

3. $a = (-5, 2, 7), b = (3, -1, 2)$

4. $a = (3, 4), b = (-4, 3)$

Proof. The projection of $\vec{b}$ onto $\vec{a}$ is:

$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

1. $\frac{6}{3}(1, -1, 1) = 2(1, -1, 1)$

2. 0

3. $-\frac{1}{26} \vec{a}$

4. 0

□

Problem 5.3.8. In n -dimensional space, let $a_1, \dots, a_k$ be vectors. The parallelotope is defined as:

$$\{t_1 a_1 + \dots + t_k a_k \mid 0 \leq t_1, \dots, t_k \leq 1\}$$

Then the area of the parallelogram spanned by a, b is:

$$\sqrt{|a|^2 |b|^2 - (\vec{a} \cdot \vec{b})^2}$$

Show that this equals $|a||b| \sin \theta$.

Proof. The area of parallelogram is a product of $|a|$ and the orthogonal component of b with respect to a . Therefore, area of the parallelogram is: $|a| \left| b - \frac{a \cdot b}{|a|^2} a \right|$. But then:

$$\begin{aligned} () |a| \left| b - \frac{a \cdot b}{|a|^2} a \right|^2 &= |a|^2 () |b|^2 - \frac{(a \cdot b)^2}{|a|^2} \\ &= |a|^2 |b|^2 - (a \cdot b)^2 \end{aligned}$$

Since $a \cdot b = |a||b| \cos \theta$, we have:

$$\begin{aligned} |a|^2 |b|^2 - (a \cdot b)^2 &= |a|^2 |b|^2 - |a|^2 |b|^2 \cos^2 \theta \\ &= |a|^2 |b|^2 (1 - \cos^2 \theta) \\ &= |a|^2 |b|^2 \sin^2 \theta \end{aligned}$$

□

Problem 5.3.9. Let P be the parallelogram formed by vectors A and B in $\mathbb{R}^3$ space. Let a_1, a_2, a_3 be the projected areas of P onto the yz -, zx -, and xy -planes respectively. Show that:

$$a^2 = a_1^2 + a_2^2 + a_3^2 \quad \text{and} \quad a_i = a \cos \theta_i \quad (i = 1, 2, 3)$$

where θ_i is the angle between P and the i -th coordinate plane. Is $(\cos \theta_1, \cos \theta_2, \cos \theta_3)$ a unit vector?

Proof. Write $A = (A_1, A_2, A_3)$ and $B = (B_1, B_2, B_3)$. Then by the previous problem,

$$\begin{aligned} a^2 &= |A|^2 |B|^2 - (A \cdot B)^2 \\ &= (A_1^2 + A_2^2 + A_3^2)(B_1^2 + B_2^2 + B_3^2) - (A_1 B_1 + A_2 B_2 + A_3 B_3)^2 \\ &= (A_2 B_3 - A_3 B_2)^2 + (A_3 B_1 - A_1 B_3)^2 + (A_1 B_2 - A_2 B_1)^2 \end{aligned}$$

On the other hand, a_1 is the area of the parallelogram formed by $(0, A_2, A_3)$ and $(0, B_2, B_3)$. Therefore,

$$a_1^2 = (A_2 B_3 - A_3 B_2)^2$$

Similarly, we can show that $a_2^2 = (A_3 B_1 - A_1 B_3)^2$ and $a_3^2 = (A_1 B_2 - A_2 B_1)^2$. Hence we get $a^2 = a_1^2 + a_2^2 + a_3^2$. Also, by the property of the projection, $a^2 = a_1^2 + a_2^2 + a_3^2 = a^2(\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3)$, which implies that $(\cos \theta_1, \cos \theta_2, \cos \theta_3)$ is a unit vector. $\square$

Problem 5.3.14. Under the constraint $x^2 + 2y^2 + 3z^2 = 1$, find the maximum and minimum of $x + y + z$, and the pair (x, y, z) that gives the maximum and minimum.

Proof. Let $\vec{a} = (x, \sqrt{2}y, \sqrt{3}z)$ and $\vec{b} = \left(1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}\right)$. Then apply the Cauchy–Schwarz inequality:

$$|\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}| \Rightarrow |x + y + z| \leq \sqrt{x^2 + 2y^2 + 3z^2} \cdot \sqrt{1 + \frac{1}{2} + \frac{1}{3}} = \sqrt{\frac{11}{6}}$$

Equality holds when $\vec{a}$ and $\vec{b}$ are in the same or opposite directions. Thus,

1. Maximum value of $x + y + z$ is obtained when $\vec{a} = \lambda \vec{b}$:

$$(x, y, z) = \left(\sqrt{\frac{6}{11}}, \frac{1}{2} \sqrt{\frac{6}{11}}, \frac{1}{3} \sqrt{\frac{6}{11}} \right)$$

2. Minimum value of $x + y + z$ is obtained when $\vec{a} = -\lambda \vec{b}$:

$$(x, y, z) = \left(-\sqrt{\frac{6}{11}}, -\frac{1}{2} \sqrt{\frac{6}{11}}, -\frac{1}{3} \sqrt{\frac{6}{11}} \right)$$

Also, the maximum value of $x + y + z$ is $\sqrt{\frac{11}{6}}$ and the minimum value is $-\sqrt{\frac{11}{6}}$.

□

9 9th week

Today, we are going to discuss about vector space, and the span of them. My reference is *Linear Algebra* by Friedberg.

9.1 Problems

Problem 5.4.5. *Let planes $x + y + z = 1$ and $x - y - z = 5$ is given.*

1. *Find the intersection of these two planes.*
2. *What is the distance between this intersection and the origin?*

Proof.

1. By substituting the second equation into the first, we get $y + z = -2$. Therefore the equation of the line we are looking for is:

$$y + z = -2, \quad x = 3.$$

2. Let p be a point on the line of (1). Then we can write $p = (3, -2 - t, t)$ for some $t \in \mathbb{R}$. Then the minimum value of $\sqrt{9 + (-2 - t)^2 + t^2}$ is the distance we are looking for. When $t = -1$, we get the minimum $\sqrt{11}$.

□

Problem 5.4.7. Calculate the reflection of $P = (1, 1, 1)$ across the plane $x + y + 2z = 1$.

Proof. Let's generalize a little bit. Suppose that Q is in the plane whose normal vector is $\mathbf{n}$. Then we can write that plane as $\mathbf{n} \cdot (X - Q) = 0$. Reflecting certain point across the plane is nothing more than just finding the coefficient of $\mathbf{n}$ which is consisting $(P - Q)$, and inverting the sign. Now write the reflection of P as P' . Then we only need to calculate the following:

$$P' - Q = (P - Q) - 2 \frac{(P - Q) \cdot \mathbf{n}}{|\mathbf{n}|^2} \mathbf{n}$$

Now let's go back to our problem. We can easily observe that the plane is given by $(x, y - 1, z) \cdot (1, 1, 2) = 0$. In this case, $Q = (0, 1, 0)$, and $\mathbf{n} = (1, 1, 2)$. Therefore, we have:

$$\begin{aligned} P' - Q &= (P - Q) - 2 \frac{(P - Q) \cdot \mathbf{n}}{|\mathbf{n}|^2} \mathbf{n} \\ &= (1 - 0, 1 - 1, 1 - 0) - 2 \frac{(1, 0, 1) \cdot (1, 1, 2)}{6} (1, 1, 2) \\ &= (1, 0, 1) - 2 \frac{3}{6} (1, 1, 2) \\ &= (1, 0, 1) - (1, 1, 2) \\ &= (0, -1, -1) \end{aligned}$$

□

Problem 5.5.2. *Determine whether the following vectors are linearly independent or linearly dependent.*

1. $\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (2, 3, 4), \quad \mathbf{c} = (7, 5, 6)$

2. $\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (2, 3, 1), \quad \mathbf{c} = (3, 1, 2)$

3. $\mathbf{a} = (0, 0, 0), \quad \mathbf{b} = (1, 1, 1), \quad \mathbf{c} = (-1, 2, -1)$

4. $\mathbf{a} = (0, 1, 1, 1), \quad \mathbf{b} = (1, 0, 1, 1), \quad \mathbf{c} = (1, 1, 0, 1), \quad \mathbf{d} = (1, 1, 1, 0)$

Proof. Every vectors are linearly independent except for (3).

□

Problem 5.5.4. When the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, show that

$$\mathbf{a} = a_1\mathbf{u} + a_2\mathbf{v} + a_3\mathbf{w}, \quad \mathbf{b} = b_1\mathbf{u} + b_2\mathbf{v} + b_3\mathbf{w}, \quad \mathbf{c} = c_1\mathbf{u} + c_2\mathbf{v} + c_3\mathbf{w}$$

are linearly independent if and only if

$$(a_1, a_2, a_3), \quad (b_1, b_2, b_3), \quad (c_1, c_2, c_3)$$

are linearly independent.

Proof. Let $(a_1, a_2, a_3) = \mathbf{p}$, $(b_1, b_2, b_3) = \mathbf{q}$, $(c_1, c_2, c_3) = \mathbf{r}$. Then, $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent if and only if

$$x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = \mathbf{0} \text{ has only the trivial solution } x = y = z = 0$$

$$\iff (a_1x + b_1y + c_1z)\mathbf{u} + (a_2x + b_2y + c_2z)\mathbf{v} + (a_3x + b_3y + c_3z)\mathbf{w} = \mathbf{0}$$

$$\iff \text{Since } \mathbf{u}, \mathbf{v}, \mathbf{w} \text{ are linearly independent, each coefficient must be zero:}$$

$$a_1x + b_1y + c_1z = 0, \quad a_2x + b_2y + c_2z = 0, \quad a_3x + b_3y + c_3z = 0$$

$$\iff x\mathbf{p} + y\mathbf{q} + z\mathbf{r} = \mathbf{0} \text{ has only the trivial solution } x = y = z = 0$$

$$\iff \mathbf{p}, \mathbf{q}, \mathbf{r} \text{ are linearly independent.}$$

□

Problem 6.1.8. Let $n \times n$ matrices A, B are given. Write j -th column of B as b_j . Show that j -th column of AB is Ab_j . That is, show that

$$AB = A \begin{pmatrix} b_1 & b_2 & \cdots & b_n \end{pmatrix} = \begin{pmatrix} Ab_1 & Ab_2 & \cdots & Ab_n \end{pmatrix}$$

Proof. Note that $(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$. Then the j -th column of AB is:

$$\begin{pmatrix} \sum_{k=1}^n a_{1k}b_{kj} \\ \sum_{k=1}^n a_{2k}b_{kj} \\ \vdots \\ \sum_{k=1}^n a_{nk}b_{kj} \end{pmatrix} = Ab_j$$

□

Problem 6.1.9 (Markov Chain). *According to a survey on migration trends, 5% of people move from rural areas to cities each year, while 2% of people move from cities to rural areas. Let the transition matrix of the population between rural and urban areas be given by*

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 0.98 & 0.02 \\ 0.05 & 0.95 \end{pmatrix}$$

That is, $p_{11} = 0.98$ represents the proportion of people who remain in rural areas next year among those who currently live in rural areas, and $p_{12} = 0.02$ represents the proportion of people who move to urban areas. Likewise, $p_{21} = 0.05$ and $p_{22} = 0.95$ represent the same for people currently living in cities.

Assuming the total population remains constant, what does each element of matrix P^n represent?

Proof. (i,j) -th element of P^n represents the proportion of people who are currently living in i or j area or transferring from i to j after n years. □

Problem 6.1.12. Let A be an $n \times n$ square matrix. We call A as a diagonal matrix if its elements are all zero except for diagonal elements. Now suppose that A, B be $n \times n$ square matrices. Show that $AB = BA$.

Proof. Note that $a_{ij} = 0$ if $i \neq j$. Therefore,

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = \begin{cases} 0 & i \neq j \\ a_{ii}b_{ii} & i = j \end{cases}$$

That is, AB is a diagonal matrix again. But then we get:

$$AB = (a_{ii}b_{ii}) = (b_{ii}a_{ii}) = BA$$

□

10 10th week

10.1 Breif Recall of the Last Week

We have studied about the matrix, and how to multiply them last time. As I briefly mentioned last week, we can compute "determinant" of a matrix. Also, matrix and its determinant has a close relation with the linear transformation. So today, we will study the definition of the linear transformation, determinant, how to calculate it, and where to use it.

Definition 10.1.1.

Let V , and W be vector spaces over $\mathbb{R}$. A function $T : V \rightarrow W$ is called a **linear transformation** if the following two conditions are satisfied:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u} \in V$ and $c \in \mathbb{R}$

The following is an useful fact which you would have learned in calculus 1 class.

Fact: There is a one to one correspondence between linear transformations and matrices.

The way to transform a linear transformation into a matrix is what we will goint to study today. Since the linear transformation is also a map, we can discuss something more about its inverse, compositin, and the condition when it is injective. To do so, we should take a look at the definition of the determinant.

Definition 10.1.2.

Given a matrix

$$A = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix},$$

the determinant of A is defined as:

$$\det(A) = \begin{vmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{vmatrix} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)}.$$

But in fact, to practically calculate the determinant, this definition is too complicated to use. But fortunaltely, we have some techniques:

10.2 Some Tecniques to Calculate Determinant

Definition 10.2.1. The followings are called to be **elementary row operations**:

1. Interchange two rows

2. *Multiply a row by a nonzero constant*
3. *Add a multiple of one row to another row*

Similarly, we can define **elementary column operations** as well.

Then we have a useful fact:

Theorem 10.2.2.

1. *First elementary row(column) operation changes the sign of determinant.*
2. *For the second elementary row(column) operation, the determinant is multiplied by the constant we multiplied.*
3. *Third elementary row(column) operation does not change the determinant.*

Also, we can calculate the determinant of a matrix by using the **Laplace expansion**:

Definition 10.2.3.

- The **minor** of the entry a_{ij} , denoted by M_{ij} , is the determinant of the submatrix obtained by deleting the i th row and j th column from A .
- The **cofactor** of the entry a_{ij} , denoted by C_{ij} , is defined as

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Theorem 10.2.4 (Laplace Expansion).

$$\det(A) = \sum_{k=1}^n a_{kj} C_{kj} = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj} \quad (\text{expansion along the } j\text{th column}),$$

$$\det(A) = \sum_{k=1}^n a_{ik} C_{ik} = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in} \quad (\text{expansion along the } i\text{th row}).$$

10.3 Relation between Determinant, Matrix, and Linear Transformation

First we define the inverse matrix:

Definition 10.3.1. Let A be a square matrix. A^{-1} is *the inverse matrix* if

$$AA^{-1} = A^{-1}A = I.$$

(In fact, one of conditions above can be omitted.)

Note that, for some matrix, there may not exist the inverse matrix of it. But the following theorem characterizes matrices which have the inverse matrices.

Theorem 10.3.2. Let A be a square matrix. The followings are equivalent:

1. A is invertible, i.e. has the inverse matrix.
2. $\det(A) \neq 0$.
3. Row vectors(or column vectors) of A are linearly independent.

You may find the proof of the theorem in the textbook. Also, we can have a new kind of matrix:

Definition 10.3.3. Let A be a square matrix, and let C_{ij} be its (i, j) -th cofactor. The **adjoint matrix** of A is:

$$\text{adj}(A) = \left((C_{ij})_{1 \leq i, j \leq n} \right)^t = (C_{ji})_{1 \leq i, j \leq n}$$

Then we can get somewhat important result:

Theorem 10.3.4. Let A be an $n \times n$ square matrix. Then the following holds:

$$A \text{adj}(A) = \det(A) I_n$$

Proof. Write $D = A \text{adj}(A)$. Then:

$$D_{ij} = \sum_{k=1}^n a_{ik} C_{jk}.$$

If $i = j$, it is $\det(A)$. If $i \neq j$, consider the matrix $A' = [a'_{ij}]$ obtained from A by replacing the j th row of A

by its i th row. That is,

$$A' = (a'_{ij}) = \begin{cases} a'_{pq} = a_{pq} & \text{if } p \neq j \\ a'_{pq} = a_{iq} & \text{if } p = j \end{cases}$$

Then A' is a matrix with two identical rows, hence its determinant is zero. Also,

$$\det(A') = \sum_{k=1}^n a_{ik} C_{jk} = D_{ij} = 0.$$

Therefore we have:

$$D_{ij} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Therefore $D = \det(A)I_n$. □

Corollary 10.3.5. *Let A be an invertible square matrix. Then we get:*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Proof. Direct from the previous theorems. □

Corollary 10.3.6. *Let A be an $n \times n$ square matrix where $n > 1$.*

- $\operatorname{adj}(\mathbf{I}_n) = \mathbf{I}_n$.
- $\operatorname{adj}(\mathbf{0}) = \mathbf{0}$.
- $\operatorname{adj}(cA) = c^{n-1} \operatorname{adj}(A)$ for any $c \in \mathbb{R}$.
- $\operatorname{adj}(A^\top) = \operatorname{adj}(A)^\top$.
- $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.
- If A is invertible, then

$$\operatorname{adj}(A) = (\det A)A^{-1}.$$

It follows that:

- $\operatorname{adj}(A)$ is invertible with inverse $(\det A)^{-1}A$.
- $\operatorname{adj}(A^{-1}) = \operatorname{adj}(A)^{-1}$.

Proof. Easy! Try this. □

Note: You may find the definition of the adjoint matrix in the page 297 in the textbook.

10.4 Problems

Problem 6.2.2. Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$L(1, -1, 1) = (2, 1), \quad L(0, 2, 1) = (1, 0), \quad L(2, 1, 2) = (0, 3).$$

Find $L(0, -1, 1)$.

Proof. Note that $2(1, -1, 1) + (0, 2, 1) - (2, 1, 2) = (0, -1, 1)$. Therefore $L(0, -1, 1) = 2L(1, -1, 1) + L(0, 2, 1) - L(2, 1, 2) = 2(2, 1) + (1, 0) - (0, 3) = (5, -1)$. $\square$

Problem 6.2.7. Let $f(P)$ be the foot of the perpendicular dropped from a point P in space to the plane $x + 2y - z = 0$.

(1) Find $f(P)$ when $P = (a, b, c)$.

(2) Show that $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a linear transformation and find the matrix corresponding to f .

Proof. Let $\mathbf{n} = (1, 2, -1)$, which is a normal vector of the given plane. Recall what we did in the problem 5.4.7. We showed that

$$f(P) = P - \frac{\mathbf{n} \cdot P}{|\mathbf{n}|^2} \mathbf{n}.$$

With some easy calculation, we can get $f(kX + Y) = kf(X) + f(Y)$, hence it is a linear transformation. Finding the image of f for the standard basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$,

$$\begin{aligned} f(\mathbf{i}) &= \mathbf{i} - p_{\mathbf{n}}(\mathbf{i}) = (1, 0, 0) - \frac{1}{6}(1, 2, -1) \\ f(\mathbf{j}) &= \mathbf{j} - p_{\mathbf{n}}(\mathbf{j}) = (0, 1, 0) - \frac{1}{6}(2, 4, -2) \\ f(\mathbf{k}) &= \mathbf{k} - p_{\mathbf{n}}(\mathbf{k}) = (0, 0, 1) - \frac{1}{6}(-1, -2, 1) \end{aligned}$$

Therefore, the matrix corresponding to f is

$$I_3 - \frac{1}{6} \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}$$

□

Problem 6.2.12. Given an $m \times n$ matrix A and an $n \times \ell$ matrix B , consider two linear transformations:

$$L_B : \mathbb{R}^\ell \rightarrow \mathbb{R}^n, \quad L_A : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Show that the composition

$$L_A \circ L_B : \mathbb{R}^\ell \rightarrow \mathbb{R}^m$$

is equal to

$$L_{AB} : \mathbb{R}^\ell \rightarrow \mathbb{R}^m.$$

That is, the product of matrices corresponds to the composition of linear transformations.

Proof.

$$\begin{aligned}(L_A \circ L_B)(X) &= L_A(L_B(X)) \\ &= A(BX) \\ &= (AB)X \\ &= L_{AB}(X).\end{aligned}$$

□

Problem 6.2.13. *Show that a necessary and sufficient condition for a linear transformation $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be one-to-one is that the equation $L(X) = O$ has only the trivial solution.*

L is one-to-one

$$\iff (L(X_1) = L(X_2)) \Rightarrow X_1 = X_2$$

$$\iff (L(X_1) - L(X_2) = L(X_1 - X_2) = 0) \Rightarrow X_1 - X_2 = 0$$

$$\iff (L(X) = 0 \Rightarrow X = 0)$$

Problem 7.1.4.

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Show that A has the inverse iff $\lambda_1 \lambda_2 \cdots \lambda_n \neq 0$, and find its inverse.

Proof. Determinant of it is $\lambda_1 \cdots \lambda_n$. Therefore it is invertible iff $\lambda_1 \cdots \lambda_n \neq 0$. Also, note that

$$\text{adj}(A) = \begin{pmatrix} \lambda_2 \cdots \lambda_n & 0 & \cdots & 0 \\ 0 & \lambda_1 \lambda_3 \cdots \lambda_n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_1 \lambda_2 \cdots \lambda_{n-1} \end{pmatrix}$$

Therefore, we have:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}$$

□

Problem 7.1.5. Find the inverse of the matrix $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$, if it exists, and use it to solve the following system of equations:

$$\begin{cases} x + 3y = 7, \\ 2x + 5y = -3 \end{cases}$$

Proof.

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{5 - 6} \begin{pmatrix} 5 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}.$$

The given equation system can be represented by:

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ -3 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \begin{pmatrix} 7 \\ -3 \end{pmatrix} = \begin{pmatrix} -44 \\ 17 \end{pmatrix}$$

□

Problem 7.1.7.

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Show that $A^2 = 2A - I$, and find the inverse of A .

Proof.

$$A^2 = \begin{pmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 2A - I$$

Note that $\det(A) = 1$, hence invertible. Then we get:

$$A^{-1}I = A^{-1}(2A - A^2) = 2I - A.$$

□

Note: I suggest you to refer the Cayley-Hamilton Theorem.

Problem 7.1.8. For an invertible $n \times n$ matrix A , show that:

$$(A^{-1})^{-1} = A, \quad (A^{-1})^k = (A^k)^{-1} \quad (k = 1, 2, 3, \dots)$$

Also show that the transpose A^t is also invertible and satisfies:

$$(A^t)^{-1} = (A^{-1})^t$$

Moreover, if B is also an invertible $n \times n$ matrix, then AB is invertible and:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof. To show that A^t and AB are invertible, just use det argument. $(A^{-1})^{-1}$ is, by the definition, the matrix such that $A^{-1}(A^{-1})^{-1} = I$. Since $A^{-1}A = I$, we are done. Similarly,

$$\begin{aligned} A^k(A^{-1})^k &= (A^{-1})^k A^k = I \quad \text{so} \quad (A^{-1})^k = (A^k)^{-1} \\ A^{-1}A &= AA^{-1} = I \quad \text{so} \quad (A^t(A^{-1})^t = (A^{-1})^t A^t = I^t = I) \\ &\Rightarrow (A^t)^{-1} = (A^{-1})^t \\ (AB)(B^{-1}A^{-1}) &= (B^{-1}A^{-1})AB = I \quad \text{so} \quad (AB)^{-1} = B^{-1}A^{-1} \end{aligned}$$

□

Problem 7.2.1.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

(1) Find the compositions $\sigma\tau$ and $\tau\sigma$.

(2) Find $\text{sgn}(\sigma\tau), \text{sgn}(\sigma), \text{sgn}(\tau), \text{sgn}(\tau^{-1})$.

Proof.

1.

$$\sigma\tau(1) = \sigma(\tau(1)) = \sigma(2) = 3,$$

$$\sigma\tau(2) = \sigma(\tau(2)) = \sigma(3) = 2,$$

$$\sigma\tau(3) = \sigma(\tau(3)) = \sigma(1) = 1$$

$$\Rightarrow \sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Similarly,

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

2.

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \quad \tau^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

Check how many transpositions are needed to express each:

$$\Rightarrow \text{sgn}(\sigma\tau) = -1, \quad \text{sgn}(\sigma) = -1, \quad \text{sgn}(\tau) = 1, \quad \text{sgn}(\tau^{-1}) = 1$$

□

Problem 7.2.2. List all 3-permutations and classify them as even or odd.

Proof.

$$\begin{aligned}\sigma_1 = \text{id} &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, & \sigma_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, & \sigma_3 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \\ \sigma_4 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, & \sigma_5 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, & \sigma_6 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}\end{aligned}$$

Here, $\sigma_1, \sigma_2, \sigma_3$ are even permutations, and $\sigma_4, \sigma_5, \sigma_6$ are odd permutations. □

11 11th week

For this week, we'll try some more problems about det, and study the vector product.

11.1 Vector Product

Definition 11.1.1. Let $\mathbf{a}$, and $\mathbf{b}$ be vectors in $\mathbb{R}^3$. The **vector product**(or **cross product**) of them is:

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}||\mathbf{b}|\sin\theta)\mathbf{n},$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $\mathbf{n}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$. Note that both of θ and $\mathbf{n}$ follow the right hand rule.

Write $\mathbf{a} = (a_1, a_2, a_3)$, and $\mathbf{b} = (b_1, b_2, b_3)$. Then equivalently, we can compute $\mathbf{a} \times \mathbf{b}$ as follows:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3 + a_1b_2 - a_2b_1)$$

Also, by the alternating property of the determinant, we can observe that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$. This is just an operation, but we can say a lot of things if we combine it with a curve in some space. We will study it in the next week.

11.2 Problems

Problem 7.3.1. *Calculate the det.*

1. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

2. $\begin{pmatrix} 1 & 9 & 99 \\ 0 & 2 & 88 \\ 0 & 0 & 3 \end{pmatrix}$

3. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$

Proof. Simple calculations with elementary row operations and Laplace expansion.

1. 0

2. 6

3. 2

□

Problem 7.3.2. Show that the area of the parallelogram formed by two vectors $\mathbf{x}, \mathbf{y}$ in the coordinate plane,

$$\{s\mathbf{x} + t\mathbf{y} : 0 \leq s, t \leq 1\}$$

is given by $|\det(\mathbf{x}, \mathbf{y})|$.

Proof. Write $\mathbf{x} = (x_1, x_2)$, and $\mathbf{y} = (y_1, y_2)$. Extend them into $\mathbb{R}^3$ vectors by $\mathbf{x}' = (x_1, x_2, 0)$, and $\mathbf{y} = (y_1, y_2, 0)$. By the definition of the cross product, we get:

$$\begin{aligned} |\mathbf{x}' \times \mathbf{y}'| &= |\mathbf{x}'||\mathbf{y}'| \sin \theta \\ &= \left| \det \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & 0 \\ y_1 & y_2 & 0 \end{pmatrix} \right| \\ &= \left| \det \begin{pmatrix} x_1 & x_2 \\ y_1 & y_2 \end{pmatrix} \right| = |\det(\mathbf{x}, \mathbf{y})|. \end{aligned}$$

□

Problem 7.3.7. Find the area of the region in the coordinate plane defined by the disk

$$x^2 + y^2 \leq 1$$

after it is transformed by the linear transformation

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Also, for a region with area A , how does its area change under this linear transformation?

Proof. This is related to the **base change** of the vector space $\mathbb{R}^2$. By the problem 7.3.2, the area of square formed by two vertical vectors changes to the region with area $|\det(L)|$, after the linear transformation. Note every region can be expressed by infinitely many squares, the area of the region becomes $|\det(L)|$ times of the original area. Therefore the answer is 6π and $6A$. $\square$

Problem 7.3.10. *Show the followings:*

$$\det \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{pmatrix} = (x-y)(y-z)(z-x)$$

$$\det \begin{pmatrix} t & x & y \\ x & t & y \\ x & y & t \end{pmatrix} = (t-x)(t-y)(t+x+y)$$

$$\det \begin{pmatrix} a & b & 0 \\ c & d & 1 \\ e & f & 0 \end{pmatrix} = -\det \begin{pmatrix} a & b \\ e & f \end{pmatrix}$$

Proof. This is also a simple calculation with Laplace expansion. But for the second one, we can view t as a variable, and x and y as constants. The result should be a cubic polynomial in t , by the definition of the det. But when we put $t = x, y$, and $-x - y$, det becomes 0. Therefore the result follows. $\square$

Problem 8.1.2. Find the equation of the line that passes through the point $(2, 1, -5)$ and is orthogonal to both vectors $(1, 2, 3)$ and $(-3, -2, 4)$.

Proof. Direction is parallel to the cross product of them. After some computations, we get:

$$\frac{x-2}{14} = -\frac{y-1}{13} = \frac{z+5}{4}$$

□

Problem 8.1.7. For a point P in coordinate space and vectors $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (3, 2, 1)$, find the area of the parallelogram defined by the expression:

$$P + t\mathbf{v} + s\mathbf{w}, \quad 0 \leq t \leq 1, \quad 0 \leq s \leq 1.$$

Proof. Same with the problem 7.3.2. Consider the plane generated by $\mathbf{v}$ and $\mathbf{w}$, and translate P to the origin. Therefore, $|\mathbf{v} \times \mathbf{w}| = 4\sqrt{6}$. □

Problem 8.1.12. Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be three vectors in $\mathbb{R}^3$ forming a parallelepiped:

$$\{r\mathbf{x} + s\mathbf{y} + t\mathbf{z} \mid 0 \leq r, s, t \leq 1\}.$$

Show that the volume is given by $|\det(\mathbf{x}, \mathbf{y}, \mathbf{z})|$.

Proof. The base area of the parallelogram is $|\mathbf{x} \times \mathbf{y}|$, and the height is

$$\left| \frac{\mathbf{x} \times \mathbf{y}}{|\mathbf{x} \times \mathbf{y}|} \cdot \mathbf{z} \right|.$$

Thus, the volume is

$$V = |\mathbf{x} \times \mathbf{y}| \cdot \left| \frac{\mathbf{x} \times \mathbf{y}}{|\mathbf{x} \times \mathbf{y}|} \cdot \mathbf{z} \right| = |(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{z}| = |\det(\mathbf{x}, \mathbf{y}, \mathbf{z})|.$$

Note: $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{w} = \det(\mathbf{x}, \mathbf{y}, \mathbf{z})$ if and only if $\mathbf{w} = \mathbf{z}$. That is, the only vector satisfying this equation is $\mathbf{z}$.

Proof is easy, just use the multilinearity of the determinant.

□

Problem 8.1.18. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be vectors in three-dimensional space. Show the following identities:

$$(a) \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

$$(b) \quad (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$$

$$(c) \quad (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$$

$$(d) \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$$

Proof.

1. coordinatewise calculation.
2. Use the anticommutativity of the cross product.
3. Observe that:

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \det \begin{pmatrix} \mathbf{a} \times \mathbf{b} \\ \mathbf{c} \\ \mathbf{d} \end{pmatrix} = ((\mathbf{a} \times \mathbf{b}) \times \mathbf{c}) \cdot \mathbf{d}$$

Now use (a).

4. Direct from (a).

□

12 12th week

12.1 Curves

From this week, we will study about curves.

Definition 12.1.1. A curve is a function $\alpha : I \rightarrow \mathbb{R}^n$ where I is an interval. We can write $X(t) = (x_1(t), x_2(t), \dots, x_n(t))$. In this case, t is a parameter of α .

Note that curves can have same shape even if they have different parameterization. Using parameterization, we can define continuous curves and differentiable curves.

Definition 12.1.2. A curve $\alpha(t) = (x_1(t), \dots, x_n(t))$ is **continuous** if $x_i(t)$ is continuous for all i . A curve $\alpha(t)$ is **differentiable** if $x_i(t)$ is differentiable for all i . In particular, if $\alpha(t)$ is differentiable, we write as:

$$\alpha'(t) = (x'_1(t), \dots, x'_n(t)).$$

This curve can be interesting if we view it as a moving point or vector. That is, we can do some operation with it. From now on, we call α as a vector valued function.

Definition 12.1.3. Let $\alpha(t)$ and $\beta(t)$ be vector valued functions.

1. $(\alpha + \beta)(t) = \alpha(t) + \beta(t)$
2. $(\alpha \cdot \beta)(t) = \alpha(t) \cdot \beta(t)$
3. $(f\alpha)(t) = f(t)\alpha(t)$
4. If $n = 3$, $(\alpha \times \beta)(t) = \alpha(t) \times \beta(t)$

Then the following theorem holds:

Theorem 12.1.4.

1. $(\alpha + \beta)'(t) = \alpha'(t) + \beta'(t)$
2. $(\alpha \cdot \beta)'(t) = \alpha'(t) \cdot \beta(t) + \alpha(t) \cdot \beta'(t)$
3. $(f\alpha)'(t) = f'(t)\alpha(t) + f(t)\alpha'(t)$
4. If $n = 3$, $(\alpha \times \beta)'(t) = \alpha'(t) \times \beta(t) + \alpha(t) \times \beta'(t)$

For a smooth curve $\alpha(t)$, we call $\alpha'(t)$ is the **velocity** or the **tangent vector** of it. Also, we can define the **speed** of the curve as $|\alpha'(t)|$. Finally, we call $\alpha''(t)$ as the **acceleration** of $\alpha(t)$. Using them, we can define the following:

Definition 12.1.5. Let $\alpha(t)$ be a curve. The **osculating plane** of it is the plane spanned by $\alpha'(t)$ and $\alpha''(t)$ containing $\alpha(t)$. In particular, we can write it as the following formula:

$$\left((x, y, z) - \alpha(t) \right) \cdot (\alpha'(t) \times \alpha''(t)) = 0.$$

Geometrically, the formula above means a plane containing the point $\alpha(t)$ and perpendicular to the vector $\alpha'(t) \times \alpha''(t)$.

Problem 9.1.4. Consider a differentiable curve $X(t) \in \mathbb{R}^n$ that does not pass through the origin.

$$\frac{d}{dt}|X(t)| = \frac{X(t) \cdot X'(t)}{|X(t)|}$$

Show the above identity. From this, explain that if $|X(t)|$ attains a minimum at $t = t_0$, the line joining the origin and $X(t_0)$ is perpendicular to the curve $X(t)$ at that point.

Proof. Note that:

$$|X(t)|^2 = X(t) \cdot X(t).$$

Therefore, if we differentiate it, we get:

$$\begin{aligned} \frac{d}{dt}|X(t)|^2 &= \frac{d}{dt}(X(t) \cdot X(t)) \\ &= 2X(t) \cdot X'(t) \\ &= 2|X(t)| \frac{d}{dt}|X(t)|. \end{aligned}$$

For the second question, use the fact that $\frac{d}{dt}|X(t)| = 0$ is necessary. □

Problem 9.1.5. In polar coordinates, a plane curve expressed as $r = r(\theta)$ can be written in Cartesian coordinates as

$$X(\theta) = (x(\theta), y(\theta)) = r(\theta)(\cos \theta, \sin \theta)$$

Show that

$$X'(\theta) = r'(\theta)(\cos \theta, \sin \theta) + r(\theta)(-\sin \theta, \cos \theta), \quad |X'| = \sqrt{(r')^2 + r^2}$$

From this, show that the cosine of the angle between the line joining the origin and the point $X(\theta_0)$, and the tangent to the curve at $X(\theta_0)$, is given by:

$$\frac{r'(\theta_0)}{\sqrt{(r'(\theta_0))^2 + r(\theta_0)^2}}$$

Also, for the logarithmic spiral given in polar coordinates as $r = e^\theta$, show that the tangent of the curve at any point becomes the line rotated by 45° from the line joining the origin and the curve point.

Proof. First equality is direct from the chain rule. Recall that:

$$\cos \phi = \frac{X(\theta) \cdot X'(\theta)}{|X(\theta)| |X'(\theta)|}.$$

Observe:

$$\begin{aligned} X(\theta) \cdot X'(\theta) &= r(\theta)r'(\theta) \\ |X(\theta)| &= r(\theta) \\ |X'(\theta)| &= \sqrt{(r')^2 + r^2} \\ \therefore \cos \phi &= \frac{r'(\theta)}{\sqrt{(r')^2 + r^2}}. \end{aligned}$$

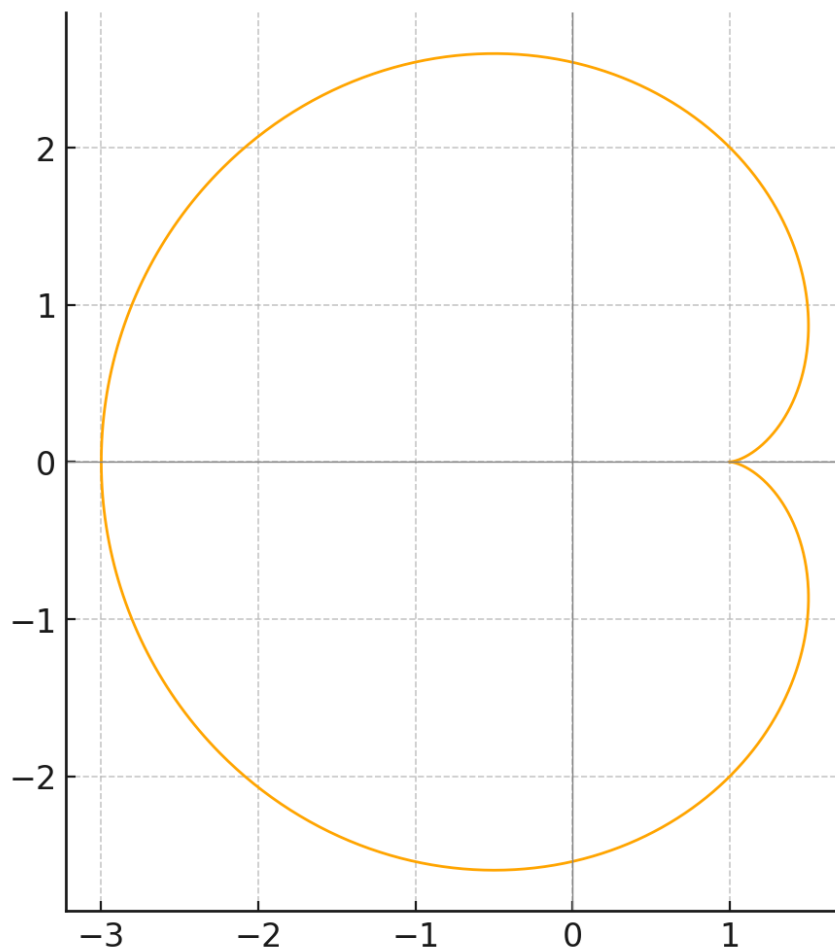
For given curve, we can parameterize it as $\alpha(\theta) = e^\theta(\cos \theta, \sin \theta)$. The remaining is just a computation. $\square$

Problem 9.1.6. In the plane, suppose point P moves as $(\cos 2t, \sin 2t)$, and point Q moves as $2(\cos t, \sin t)$. Describe the motion $X(t)$ of point Q as observed from point P , and sketch the trajectory.

Proof. At time t , let the positions of points P and Q be $P(t)$ and $Q(t)$, respectively. Then,

$$X(t) = Q(t) - P(t) = (2 \cos t - \cos 2t, 2 \sin t - \sin 2t)$$

□



Problem 9.2.1. Find the velocity and acceleration vectors of the following curves and determine the tangent plane at the given point.

1. $(e^t, \cos t, \sin t), t = 0$

2. $(\sin 2t, \log(1+t), t), t = 0$

3. $(\sin t, \cos t, t), t = \frac{\pi}{2}$

4. $(t, t^2, t^3), t = 1$

Proof.

1. • Velocity vector: $(1, 0, 1)$
 • Acceleration vector: $(1, -1, 0)$
 • Tangent plane:

$$\begin{aligned} ((x, y, z) - (1, 1, 0)) \cdot ((1, 0, 1) \times (1, -1, 0)) &= 0 \\ (x-1, y-1, z) \cdot (1, 1, -1) &= 0 \Rightarrow x + y - z - 2 = 0 \end{aligned}$$

2. • Velocity vector: $(2, 1, 1)$
 • Acceleration vector: $(0, -1, 0)$
 • Tangent plane:

$$\begin{aligned} ((x, y, z) - (0, 0, 0)) \cdot ((2, 1, 1) \times (0, -1, 0)) &= 0 \\ (x, y, z) \cdot (1, 0, -2) &= 0 \Rightarrow x - 2z = 0 \end{aligned}$$

3. • Velocity vector: $(0, -1, 1)$
 • Acceleration vector: $(-1, 0, 0)$
 • Tangent plane:

$$\begin{aligned} ((x, y, z) - (1, 0, \frac{\pi}{2})) \cdot ((0, -1, 1) \times (-1, 0, 0)) &= 0 \\ (x-1, y, z - \frac{\pi}{2}) \cdot (0, -1, -1) &= 0 \Rightarrow y + z - \frac{\pi}{2} = 0 \end{aligned}$$

4. • Velocity vector: $(1, 2, 3)$
 • Acceleration vector: $(0, 2, 6)$
 • Tangent plane:

$$\begin{aligned} ((x, y, z) - (1, 1, 1)) \cdot ((1, 2, 3) \times (0, 2, 6)) &= 0 \\ (x-1, y-1, z-1) \cdot (6, -6, 2) &= 0 \Rightarrow 3x - 3y + z - 1 = 0 \end{aligned}$$

□

Problem 9.2.2. Given the curve $X(t) = r(t)(\cos \theta(t), \sin \theta(t))$, show that

$$X' = r'(\cos \theta, \sin \theta) + r\theta'(-\sin \theta, \cos \theta)$$

$$X'' = (r'' - r(\theta')^2)(\cos \theta, \sin \theta) + \frac{1}{r}(r^2\theta')'(-\sin \theta, \cos \theta)$$

Proof. Direct from the chain rule.

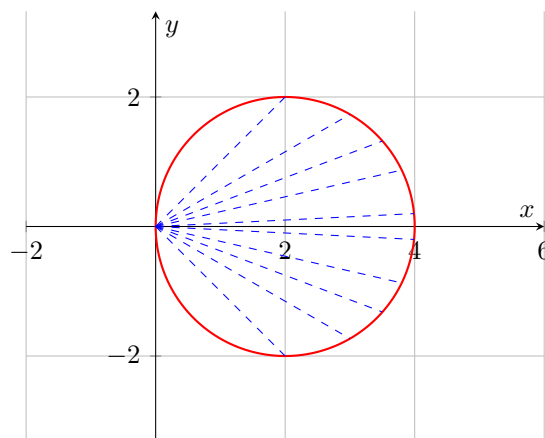
□

13 13th week

Continue studying curves. We will study about the area of the polar curve, and the reparametrization of the curve.

13.1 Area

Think about a lot of lines connecting the origin and a point of a curve. See the following figure:

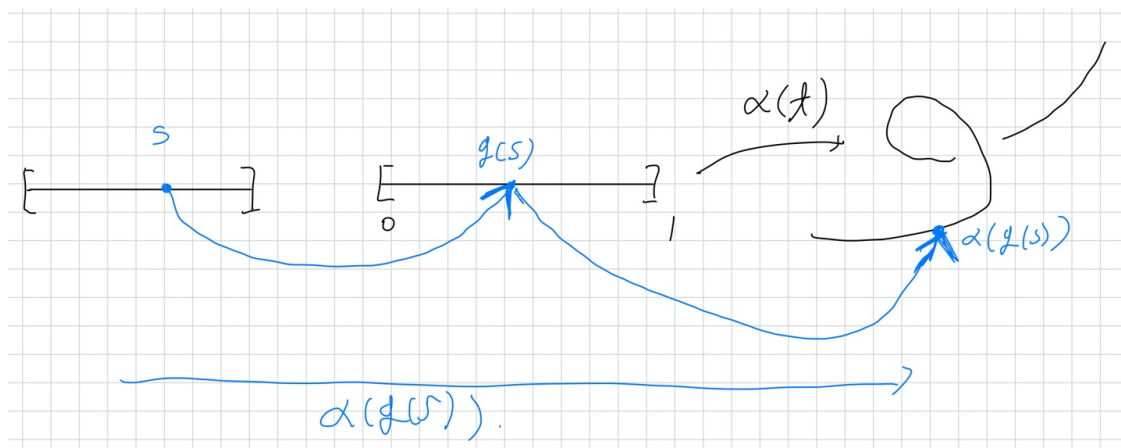


As we can observe, to calculate the area of the closed polar curve, we need to add up all the small sectors. That is, we need to integrate $\frac{1}{2}r^2$ by θ . Be careful with the sign of r . Therefore, if $r \geq 0$, we can calculate the area as:

$$\iint 1 \, dx \, dy = \int_{\theta_0}^{\theta_1} \frac{1}{2} r^2 \, d\theta$$

13.2 Reparametrization

Maybe you might be feel hard with this. Intuitively, you may think it as a pre-composition of some “good function” whose domain and codomain are both intervals.



Rigorously, let the curve $\alpha : I \rightarrow \mathbb{R}^3$ be given. If $g : J \rightarrow I$ is invertible and differentiable, we can say that $\alpha \circ g : J \rightarrow \mathbb{R}^3$ is a **reparametrization** of α . Note that the chain rule holds, i.e. $(X \circ g)'(t) = X'(g(t))g'(t)$.

13.3 Problems

Problem 9.3.3. Draw the graphs of the following polar equations, and compute the area enclosed by each curve.

1. $r = 2 \cos \theta \quad (0 \leq \theta \leq \pi)$

2. $r = \cos 3\theta$

3. $r = 1 + \sin 2\theta$

4. $r = 2 + \sin 2\theta$

5. $r^2 = \sin^2 2\theta \quad (0 \leq \theta \leq \pi)$

6. $r = \sqrt{1 - \cos \theta}$

7. $r = \sqrt{\cos 2\theta} \quad \left(-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}\right)$

8. $r = \tan\left(\frac{\pi}{4} - 2|\theta|\right) \quad \left(-\frac{\pi}{8} \leq \theta \leq \frac{\pi}{8}\right)$

Proof. Easy calculation. The result is as follows:

1. π

2. $\frac{\pi}{4}$

3. $\frac{3\pi}{2}$

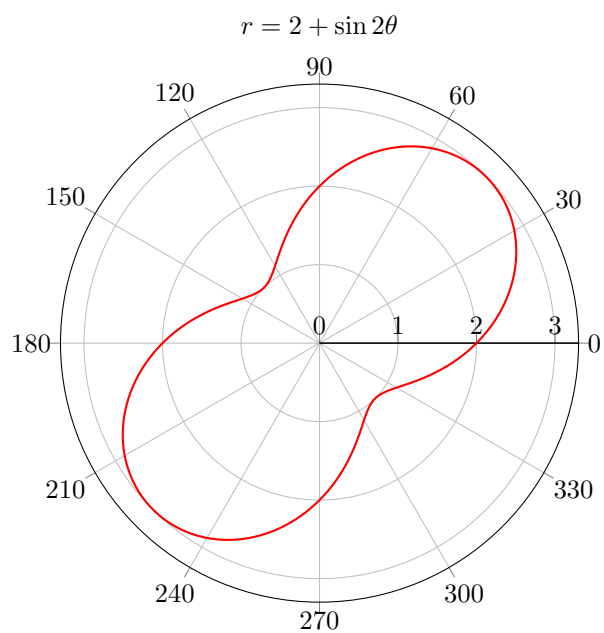
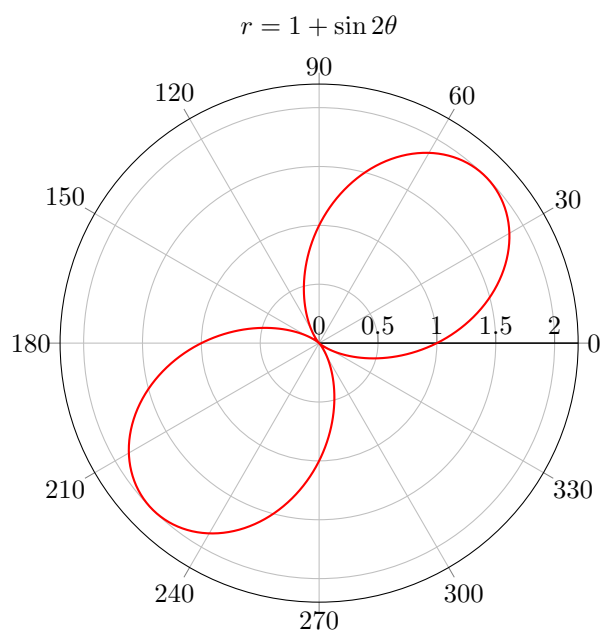
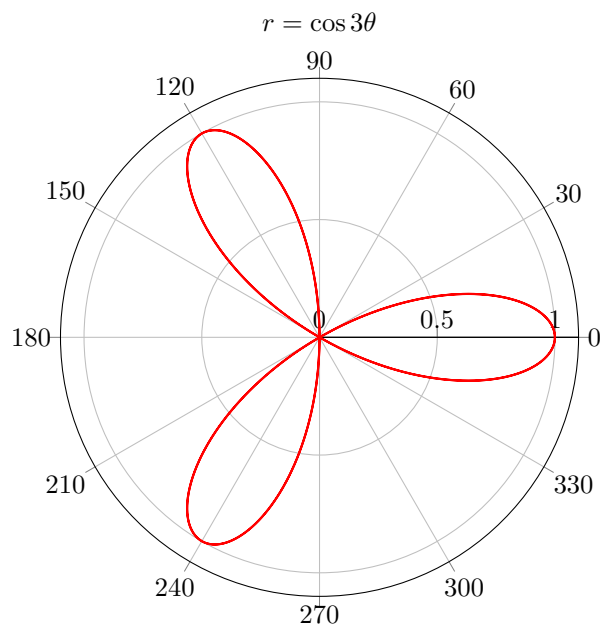
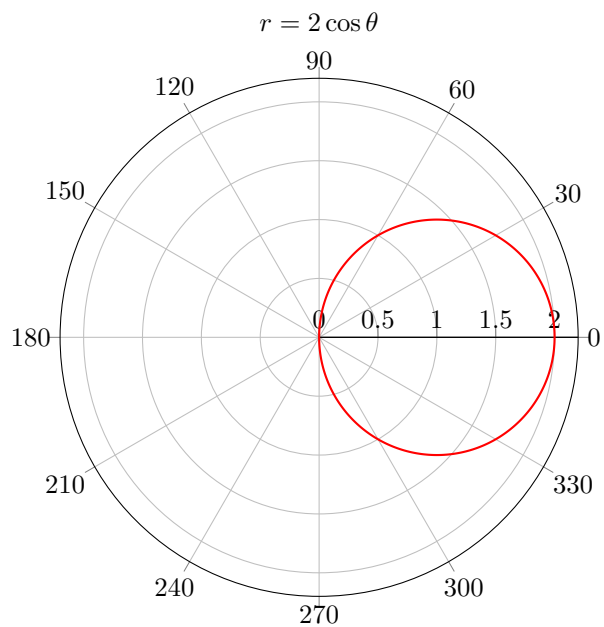
4. $\frac{9\pi}{2}$

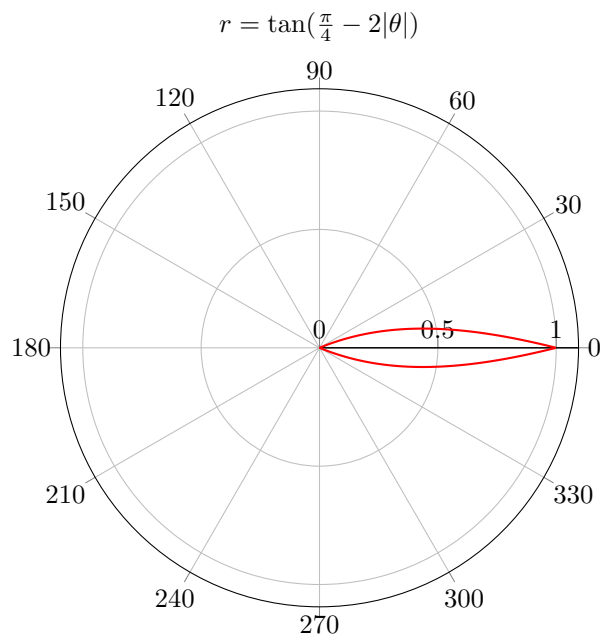
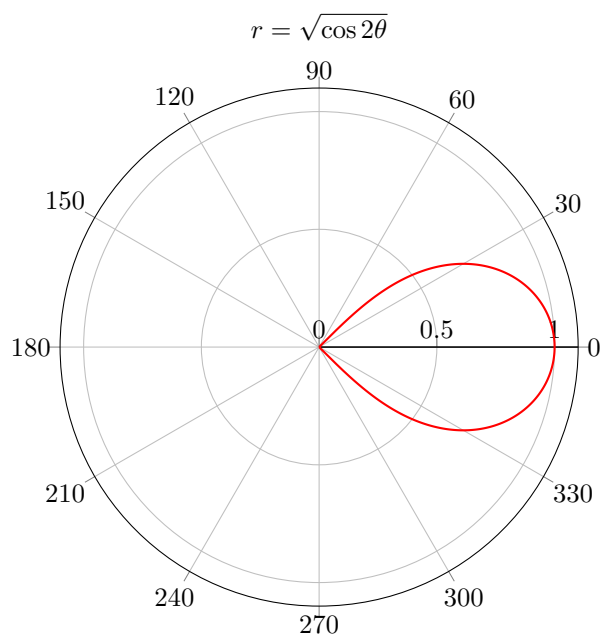
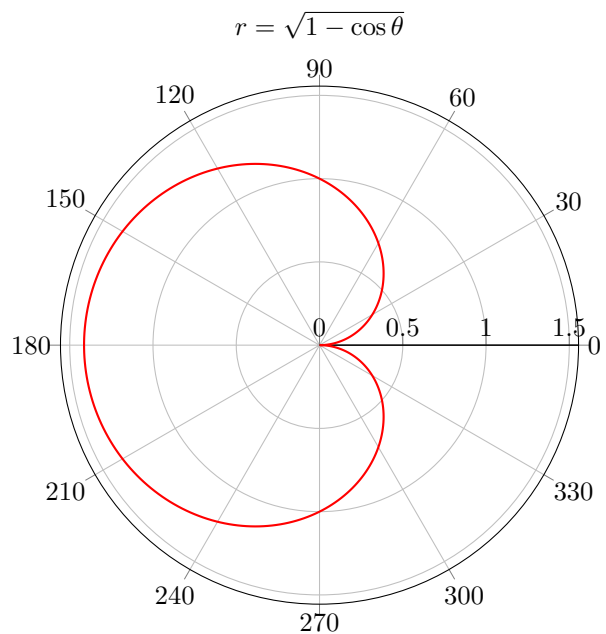
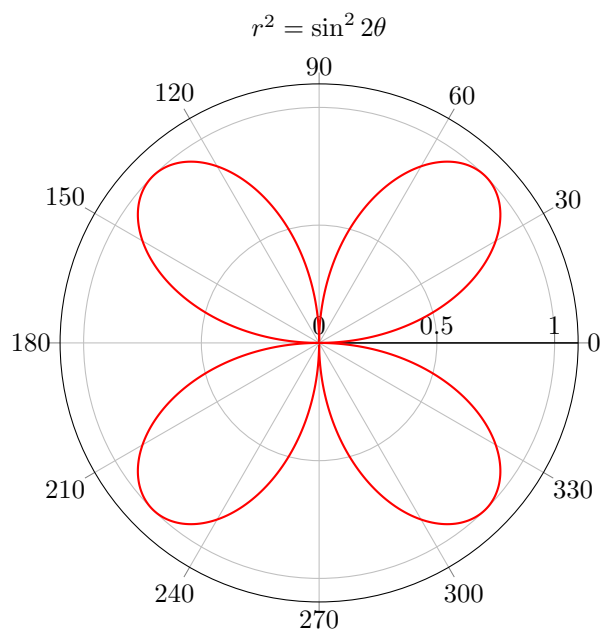
5. $\frac{\pi}{2}$

6. $\frac{\pi}{2}$

7. $\int_0^{\pi/8} \tan^2\left(\frac{\pi}{4} - 2\theta\right) d\theta = \frac{1}{2} \int_0^{\pi/4} \tan^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/4} (\sec^2 \theta - 1) d\theta = \frac{1}{2} - \frac{\pi}{8}$

□





Problem 9.3.5. Represent the polar curve given by $r = \frac{3}{2+\cos\theta}$ in Cartesian coordinates, and compute the value of

$$\int_0^{2\pi} \frac{1}{(2+\cos\theta)^2} d\theta.$$

Proof. Multiply both sides by $2 + \cos\theta$ ($\neq 0$):

$$r(2 + \cos\theta) = 3 \Rightarrow 2r + r\cos\theta = 3.$$

Convert to Cartesian coordinates:

$$2\sqrt{x^2 + y^2} + x = 3 \quad \left(\because r = \sqrt{x^2 + y^2}, \cos\theta = \frac{x}{r} \right).$$

Squaring both sides:

$$4(x^2 + y^2) = (3 - x)^2 = 9 - 6x + x^2,$$

$$3x^2 + 4y^2 + 6x - 9 = 0,$$

$$3(x+1)^2 + 4y^2 = 12 \Rightarrow \frac{(x+1)^2}{4} + \frac{y^2}{3} = 1,$$

which is an ellipse.

We know the area of ellipse is $2\sqrt{3}\pi$. Then,

$$\int_0^{2\pi} \frac{1}{(2+\cos\theta)^2} d\theta = \frac{2}{9} \int_0^{2\pi} \frac{1}{2} r^2 d\theta = \frac{4\sqrt{3}\pi}{9}.$$

□

Problem 9.3.7. For a natural number n , compute the area of a single petal of the polar curve $r = \sin n\theta$.

Proof.

$$\int_0^{\frac{\pi}{n}} \frac{1}{2} \sin^2 n\theta \, d\theta = \frac{\pi}{4n}.$$

□

Problem 9.4.1. *Parametrize the following curves.*

1. $4x^2 + 9y^2 = 1$

2. $4x^2 - 9y^2 = 1$

3. $y^2 = x^3 - x^2 \quad (x > 0)$

Proof.

1. $\left(\frac{\cos t}{2}, \frac{\sin t}{3}\right) \quad (t \in \mathbb{R})$

2. $\left(\frac{\cosh t}{2}, \frac{\sinh t}{3}\right), \left(-\frac{\cosh t}{2}, \frac{\sinh t}{3}\right) \quad (t \in \mathbb{R})$

3. $(1 + t^2, t^2(1 + t^2)) \quad (t \in \mathbb{R})$. Take $y = tx$.

□

Tip) If you see quadratic equation whose sign of variables are equal, try sin and cos. If the signs are different, try sinh and cosh. If polynomial-look like function is given, try to introduce $y = tx$.

Problem 9.4.2. Parametrize the curves described by the following equations.

1. $x^2 = 3y, \quad z = xy$

2. $x^2 - 4y^2 = 1, \quad x + y + z = 1$

3. $x^2 + y^2 + z^2 = 1, \quad x + y + z = \frac{1}{\sqrt{2}}$

Proof.

1. Let $x = t$. Then we get

$$(t^2, \frac{1}{3}t^2, \frac{1}{3}t^3).$$

2. Let $x = \cosh t, \quad y = \frac{1}{2} \sinh t$. Then,

$$z = 1 - x - y = 1 - \cosh t - \frac{1}{2} \sinh t.$$

Hence, the curve is

$$\left(\cosh t, \frac{1}{2} \sinh t, 1 - \cosh t - \frac{1}{2} \sinh t \right), \quad (t \in \mathbb{R})$$

3. Since it is the intersection of a sphere and a plane, the curve is a circle. By the symmetry of the coordinates x, y, z , the point $\frac{1}{3\sqrt{2}}(1, 1, 1)$ on the plane $x = y = z = \frac{1}{3\sqrt{2}}$ becomes the center of the circle. Let us consider the normal vector of the plane $(1, 1, 1)$ and a perpendicular vector $(2, -1, -1)$, and then compute a vector perpendicular to both:

$$(1, 1, 1) \times (2, -1, -1) = (0, 3, -3)$$

When we move the center $\frac{1}{3\sqrt{2}}(1, 1, 1)$ along the directions $(2, -1, -1)$ and $(0, 3, -3)$, which lie in the plane, let us find the intersection points with the sphere, denoted P and Q . For some real numbers p and q ,

$$\left| \frac{1}{3\sqrt{2}}(1, 1, 1) + p(2, -1, -1) \right|^2 = 1$$

$$4p^2 + \frac{2\sqrt{2}}{3}p + \frac{1}{18} + 2p^2 - \frac{2\sqrt{2}}{3}p + \frac{1}{9} = 1$$

$$6p^2 = 1 - \frac{1}{6} \Rightarrow p = \pm \frac{\sqrt{5}}{6}$$

Also, by the same method, $q = \pm \frac{\sqrt{5}}{6\sqrt{3}}$. Therefore, the found intersection circle for $0 \leq \theta \leq 2\pi$ is given by:

$$\frac{1}{3\sqrt{2}}(1, 1, 1) + \frac{\sqrt{5}}{6} \cos \theta (2, -1, -1) + \frac{\sqrt{5}}{5\sqrt{3}} \sin \theta (0, 3, -3)$$



Problem 9.4.4. The energy of a curve $X : [a, b] \rightarrow \mathbb{R}^n$ is defined as

$$E(X) := \frac{1}{2} \int_a^b |X'(t)|^2 dt$$

1. Compute the energy of the following curve.

$$X(t) = (t - \sin t, 1 - \cos t) \quad (0 \leq t \leq 2\pi)$$

2. Compute the energy of the following two curves and compare.

$$X(t) = (t, t^2) \quad (0 \leq t \leq 1), \quad Y(s) = (s^2, s^4) \quad (0 \leq s \leq 1)$$

Proof.

1.

$$\begin{aligned} E(X) &= \frac{1}{2} \int_0^{2\pi} |X'(t)|^2 dt \\ &= \frac{1}{2} \int_0^{2\pi} |(1 - \cos t, \sin t)|^2 dt \\ &= \frac{1}{2} \int_0^{2\pi} (2 - 2 \cos t) dt \\ &= \frac{1}{2} \cdot 4\pi = 2\pi \end{aligned}$$

2.

$$\begin{aligned} E(X) &= \frac{1}{2} \int_0^1 |X'(t)|^2 dt = \frac{1}{2} \int_0^1 (1 + 4t^2) dt = \frac{1}{2} \left(1 + \frac{4}{3}\right) = \frac{7}{6} \\ E(Y) &= \frac{1}{2} \int_0^1 |Y'(s)|^2 ds = \frac{1}{2} \int_0^1 (4s^2 + 16s^6) ds = \frac{1}{2} \left(\frac{4}{3} + \frac{16}{7}\right) = \frac{38}{21} \end{aligned}$$

Thus, $E(X) \neq E(Y)$.

□

Problem 9.5.1. For a positive number a , find the length of the curve

$$X(t) = (t \cos t, t \sin t, t), \quad 0 \leq t \leq \sqrt{2} \sinh a$$

Proof. As we did in the physics, we suffice to compute the integral of the size of the velocity of the curve.

$$|X'(t)| = \sqrt{t^2 + 2} \Rightarrow \text{Length of the curve is } \int_0^{\sqrt{2} \sinh a} \sqrt{t^2 + 2} dt = \frac{1}{2} \sinh(2a) + a = \frac{e^{2a} - e^{-2a}}{4} + a$$

□

Problem 9.5.3. For each of the following polar curves, draw their shapes and compute their lengths.

1. $r = 3\theta^2 \quad (1 \leq \theta \leq 2)$

2. $r = e^{-4\theta} \quad (1 \leq \theta \leq 2)$

3. $r = \frac{2}{\theta} \quad (1 \leq \theta \leq \sqrt{3})$

4. $r = \sin \frac{\theta}{2} \quad (0 \leq \theta \leq \pi)$

5. $r = \frac{\sqrt{3}}{2} + \cos \frac{\theta}{2}$

Proof. Use the formula we showed in the problem 9.2.2.

$$l = \int_0^\theta \sqrt{r(\theta)^2 + r'(\theta)^2} d\theta$$

1. $\int_1^2 3\theta\sqrt{\theta^2+4} d\theta = \left[(\theta^2+4)^{3/2}\right]_1^2 = 16\sqrt{2} - 5\sqrt{5}$

2. $\int_1^2 \sqrt{17}e^{-4\theta} d\theta = \frac{\sqrt{17}}{4}(e^{-4} - e^{-8})$

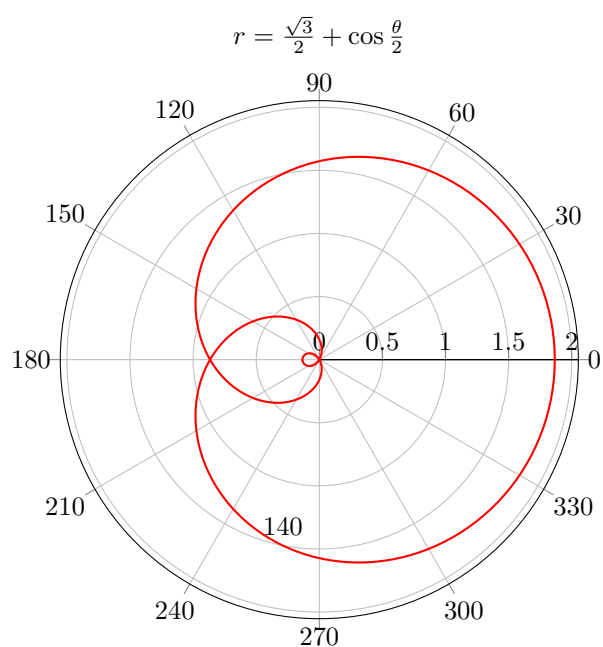
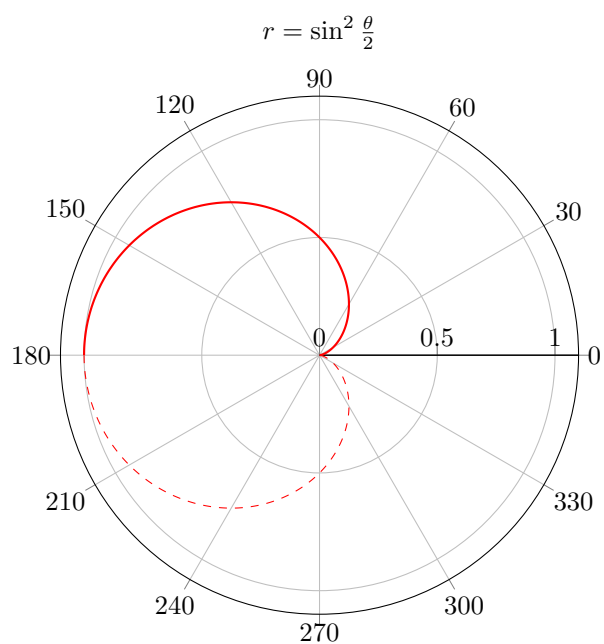
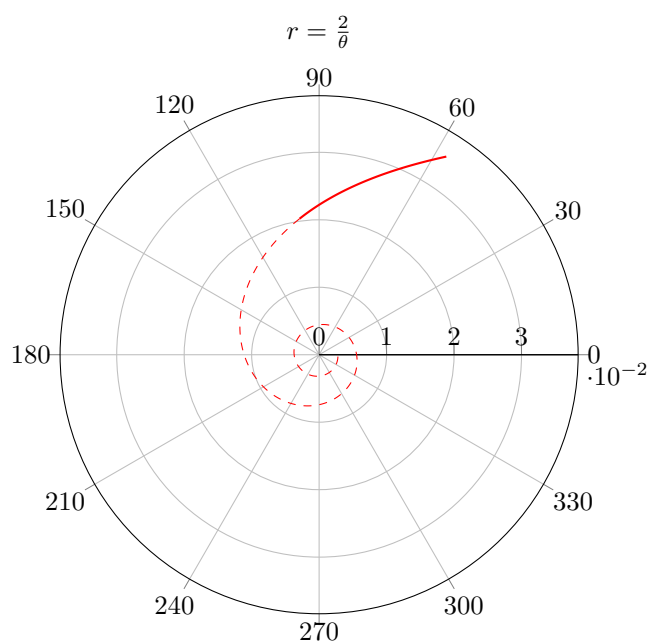
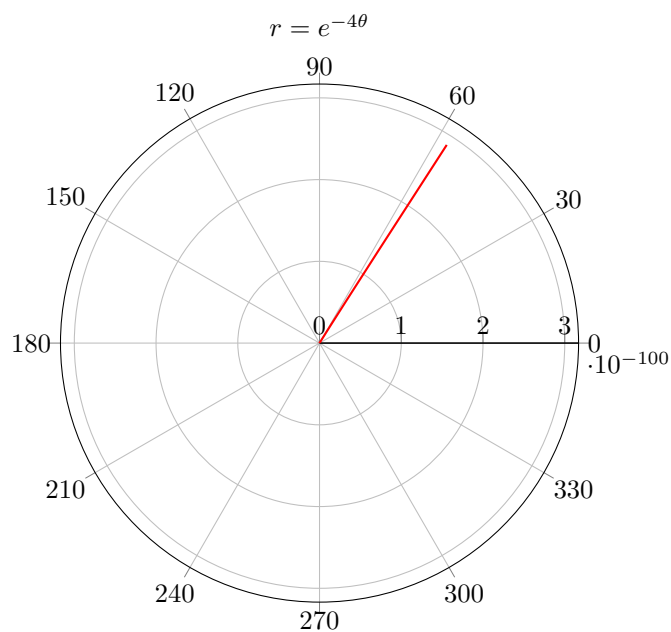
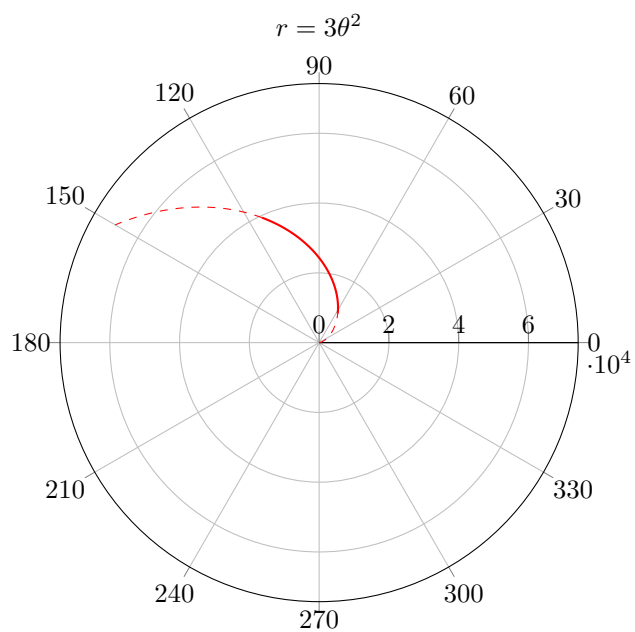
3. $\int_1^{\sqrt{3}} \frac{2\sqrt{\theta^2+1}}{\theta^2} d\theta = \int_{\pi/4}^{\pi/3} \frac{2}{\sin^2 u \cos u} du = \int_{1/\sqrt{2}}^{\sqrt{3}/2} \frac{2}{x^2(1-x^2)} dx$
 $= \left[\frac{-2}{x} + \log \left| \frac{1+x}{1-x} \right| \right]_{1/\sqrt{2}}^{\sqrt{3}/2} = 2\sqrt{2} - \frac{4}{\sqrt{3}} + \log(7+4\sqrt{3})(3-2\sqrt{2})$

In the middle, substitution $\theta = \tan u$ and $\sin u = x$ was used. Also substituting $\sinh u$ can be useful.

4. $\int_0^\pi \sin \frac{\theta}{2} d\theta = 2$

5. $\int_0^{4\pi} \left(1 + \frac{\sqrt{3}}{2} \cos \frac{\theta}{2}\right) d\theta = 4\pi \quad (\text{the range of } \theta \text{ is } 0 \leq \theta \leq 4\pi)$

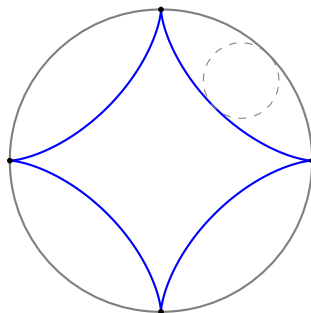
□



Problem 9.5.5. Find the length of the curve

$$\left(x^{2/3} + y^{2/3}\right) = 1.$$

This curve is the locus of a point on a small circle rolling inside a large circle. What is the ratio of the diameters of the large and small circles?



Proof. If we parametrize the astroid as $X(t) = (\cos^3 t, \sin^3 t)$ ($0 \leq t \leq 2\pi$), then $|X'(t)| = 3|\cos t \sin t|$. Thus, the length of the curve is

$$\int_0^{2\pi} 3|\cos t \sin t| dt = 6.$$

The astroid meets the large circle four times, so the ratio of diameters between the large and small circle must be 4 : 1.

□

Problem 9.5.6. For positive numbers $a < b$, find the length of the curve

$$y = \log x \quad (a \leq x \leq b).$$

Proof.

$$\begin{aligned} \int \sqrt{1 + \frac{1}{x^2}} dx &= \int \frac{\sqrt{x^2 + 1}}{x} dx && \text{put } x = \tan u \\ &= \int \frac{1}{\sin u \cos^2 u} du && \text{put } \cos u = y \\ &= \int \frac{1}{y^2(1 - y^2)} dy \\ &= -\frac{1}{y} + \frac{1}{2} \log \left| \frac{1 + y}{1 - y} \right| \\ &= \sqrt{1 + x^2} + \frac{1}{2} \log \left| \frac{\sqrt{1 + x^2} - 1}{\sqrt{1 + x^2} + 1} \right| + C. \end{aligned}$$

Put a and b , and subtract. □

14 14th week

14.1 Problems

Problem 9.6.1. *Parametrize the curve $X(t) = (t - \sin t, 1 - \cos t)$ ($0 \leq t \leq 2\pi$) in terms of arc length s .*

Proof.

$$s = \int_0^t |X'(t)| dt = \int_0^t 2 \left| \sin \frac{u}{2} \right| du = 4 \left(1 - \cos \frac{t}{2} \right) \Rightarrow t = 2 \cos^{-1} \left(1 - \frac{s}{4} \right)$$

Therefore,

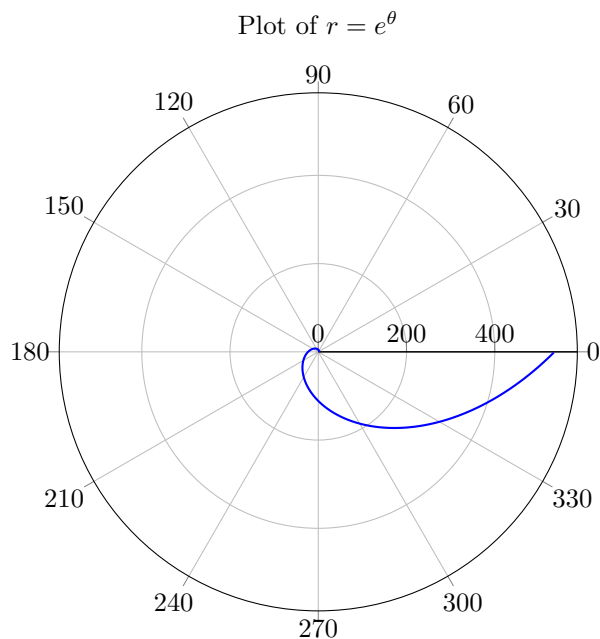
$$\begin{aligned} X(t) &= \left(2 \cos^{-1} \left(1 - \frac{s}{4} \right) - \sin \left(2 \cos^{-1} \left(1 - \frac{s}{4} \right) \right), 1 - \cos \left(2 \cos^{-1} \left(1 - \frac{s}{4} \right) \right) \right) \\ &= \left(2 \cos^{-1} \left(1 - \frac{s}{4} \right) - \frac{4-s}{8} \sqrt{8s-s^2}, \frac{8s-s^2}{8} \right) \end{aligned}$$

□

Problem 9.6.3. For the curve $X(t) = e^t(\cos t, \sin t)$ ($0 \leq t \leq 2\pi$), answer the following:

1. Sketch the curve and compute its length.
2. Reparametrize the curve in terms of arc length s measured from $X(0)$.

Proof.



1. The length is

$$\int_0^{2\pi} \sqrt{2}e^t dt = \sqrt{2}(e^{2\pi} - 1)$$

- 2.

$$s = \int_0^t \sqrt{2}e^u du = \sqrt{2}(e^t - 1) \Rightarrow t = \log\left(\frac{s}{\sqrt{2}} + 1\right)$$

Therefore, the reparametrization is:

$$X(t) = \left(\frac{s}{\sqrt{2}} + 1\right) \left(\cos \log\left(\frac{s}{\sqrt{2}} + 1\right), \sin \log\left(\frac{s}{\sqrt{2}} + 1\right)\right)$$

□

Problem 9.6.4. *Parametrize the curve $y^3 = x^2$ by arc length s .*

Proof. Let us parametrize the curve as $X(t) = (t^3, t^2)$ for $t > 0$.

$$s = \int_0^t \sqrt{(3u^2)^2 + (2u)^2} du = \int_0^t u \sqrt{9u^2 + 4} du = \frac{1}{27} \left[(9u^2 + 4)^{3/2} \right]_0^t = \frac{1}{27} (9t^2 + 4)^{3/2} - \frac{8}{27}$$

Therefore,

$$27s + 8 = (9t^2 + 4)^{3/2} \Rightarrow t = \sqrt{\frac{(27s + 8)^{2/3} - 4}{9}}$$

□

Problem 9.7.2. Evaluate the following line integrals over the given space curves $X(t)$:

1. $f(x, y, z) = xy + y - z, \quad X(t) = (2t, t, 2 - 2t), \quad 0 \leq t \leq 1$

2. $f(x, y, z) = \sqrt{x^2 + y^2}, \quad X(t) = (4 \cos t, 4 \sin t, 3t), \quad -2\pi \leq t \leq 2\pi$

3. $f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}, \quad X(t) = (t, t, t), \quad 1 \leq t$

Proof.

1. $\int_0^1 f(X(t)) |X'(t)| dt = \frac{1}{2}$

2. $\int_{-2\pi}^{2\pi} f(X(t)) |X'(t)| dt = 80\pi$

3. $\int_1^\infty f(X(t)) |X'(t)| dt = \frac{1}{\sqrt{3}}$

□

Problem 9.7.4. Find the centroid of the curve

$$x^{2/3} + y^{2/3} = 1, \quad x \geq 0, \quad y \geq 0$$

Proof. Using the parametrization $X'(t) = (\cos^3 t, \sin^3 t)$, we get $|X'(t)| = 3 \cos t \sin t$. Then the arc length is $\frac{3}{2}$ and:

$$\bar{x} = \frac{1}{3/2} \int_0^{\pi/2} \cos^3 t \cdot 3 \cos t \sin t \, dt = \frac{2}{5}$$

$$\bar{y} = \frac{1}{3/2} \int_0^{\pi/2} \sin^3 t \cdot 3 \cos t \sin t \, dt = \frac{2}{5}$$

□

Problem 9.7.5. For a line segment AB , define a point on it as $P = (1 - t)A + tB$ and a linear density function $(1 - t)a + tb$. Then, the center of mass is:

$$\frac{2a + b}{3(a + b)}A + \frac{a + 2b}{3(a + b)}B, \quad \text{i.e.,} \quad \frac{2}{3} \cdot \frac{A + B}{2} + \frac{1}{3} \left(\frac{a}{a + b}A + \frac{b}{a + b}B \right)$$

Proof. Let $X(t) = (1 - t)A + tB$, then the total mass is:

$$\int_0^1 \{(1 - t)a + tb\} |X'(t)| dt = \frac{a + b}{2} |B - A|$$

Then the center of mass is:

$$\begin{aligned} & \frac{2}{(a + b)|B - A|} \int_0^1 X(t) \cdot \{(1 - t)a + tb\} |X'(t)| dt \\ &= \frac{2}{a + b} \int_0^1 \{(1 - t)^2 aA + t(1 - t)(bA + aB) + t^2 bB\} dt \end{aligned}$$

The only remaining is just a computation. □

Problem 9.7.6. Given the space curve $X(t) = (\cos t, \sin t, t)$ and the density function $f(t) = t^2$, find the total mass and the center of mass of the curve on the interval $-\pi \leq t \leq \pi$.

Proof. Since $|X'(t)| = \sqrt{2}$, the total mass is

$$\int_{-\pi}^{\pi} t^2 \sqrt{2} dt = \frac{2\sqrt{2}\pi^3}{3}$$

The coordinates of the center of mass are:

$$\bar{x} = \frac{3}{2\sqrt{2}\pi^3} \int_{-\pi}^{\pi} \cos t \cdot t^2 \sqrt{2} dt = -\frac{6}{\pi^2}$$

$$\bar{y} = \frac{3}{2\sqrt{2}\pi^3} \int_{-\pi}^{\pi} \sin t \cdot t^2 \sqrt{2} dt = 0$$

$$\bar{z} = \frac{3}{2\sqrt{2}\pi^3} \int_{-\pi}^{\pi} t \cdot t^2 \sqrt{2} dt = 0$$

□

Problem 9.8.3. For the curve $X(t) = (t, t^2, t^3)$, find the curvature at $t = 1$ and $t = -1$.

Proof. By direct calculation using the curvature formula, the curvature at both $t = 1$ and $t = -1$ is:

$$\kappa = \frac{\sqrt{266}}{98}.$$

Tip) Use the next problem!

□

Problem 9.8.4. For a regular curve $X(t) \in \mathbb{R}^3$, using

$$\frac{d}{dt}|X'(t)| = \frac{X'(t) \cdot X''(t)}{|X'(t)|},$$

show that

$$\kappa = \frac{\sqrt{|X'|^2|X''|^2 - (X' \cdot X'')^2}}{|X'|^3} = \frac{|X' \times X''|}{|X'|^3}.$$

Also, when the angle between X' and X'' is α , show:

$$\kappa = \frac{|X''| \sin \alpha}{|X'|^2}.$$

Proof. For the first part, see the problem 9.4.1. Recall the definition of curvature:

$$\kappa(s) = |k(s)| = \frac{d}{ds} \left(\frac{X'(t)}{|X'(t)|} \right)$$

where s is the arc length parameter. Then:

$$\begin{aligned} k(s) &= \frac{d}{ds} \left(\frac{X'(t)}{|X'(t)|} \right) = \frac{dt}{ds} \frac{d}{dt} \left(\frac{X'(t)}{|X'(t)|} \right) \\ &= \frac{1}{|X'(t)|} \frac{d}{dt} \left(\frac{X'(t)}{|X'(t)|} \right) \\ &= \frac{1}{|X'(t)|} \left(-\frac{X' \cdot X''}{|X'|^3} X' + \frac{X''}{|X'|} \right) \\ &= \frac{1}{|X'|^4} ((X' \cdot X')X'' - (X' \cdot X'')X') \end{aligned}$$

Recall the basic property of the vector product: $|a \times b|^2 = |a|^2|b|^2 - (a \cdot b)^2$. Then:

$$\begin{aligned} \kappa(s) &= |k(s)| = \frac{1}{|X'|^4} |(X' \cdot X')X'' - (X' \cdot X'')X'| \\ &= \frac{1}{|X'|^4} \sqrt{|X'|^4|X''|^2 - |X'|^2(X' \cdot X'')^2} \\ &= \frac{|X' \times X''|}{|X'|^3} \end{aligned}$$

The remaining is trivial. □

Problem 9.8.6. Suppose that the cycloid is given by:

$$x(t) = t - \sin t, \quad y(t) = 1 - \cos t = 2 \sin^2 \frac{t}{2} \quad (0 \leq t \leq 2\pi)$$

Show that the curvature is:

$$\kappa(t) = \frac{1}{4 \sin \frac{t}{2}} = \frac{\sqrt{2}}{4\sqrt{y}}$$

Proof. Use the problem 4. □

Problem 9.8.12. *Given the space curve*

$$X(t) = \left(\cos t, \sin^2 t, \frac{1}{2} \sin 2t \right),$$

find the curvature and the equation of the osculating plane at $t = \pi$.

Proof.

$$X'(t) = (-\sin t, 2 \sin t \cos t, \cos 2t)$$

$$X''(t) = (-\cos t, 2 \cos 2t, -2 \sin 2t)$$

$$\therefore |X'(\pi)| = |(0, 0, -1)| = 1, \quad X''(\pi) = (-1, 2, 0)$$

$$\therefore X' \times X''(\pi) = (2, -1, 0), \text{ hence } \kappa(\pi) = \sqrt{5}, \text{ and the osculating plane is: } (2, -1, 0) \cdot (x - 1, y, z) = 0.$$

□